

IMPERIAL

HIGHER ORDER INTERACTIONS FOR CAUSAL DISCOVERY

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Abstract

Understanding higher-order interactions and causal relationships from observational data represents a fundamental challenge in modern data science. This thesis addresses two complementary objectives in causal discovery: characterizing the theoretical limits of computationally efficient unconditional tests, and establishing connections between Partial Information Decomposition (PID) and causal structure identification. We provide theoretical foundations for unconditional causal discovery by analyzing three-variable structures and introducing unconditional independence equivalence classes that precisely characterize which causal roles can be distinguished using unconditional tests alone. This leads to a systematic procedure leveraging Bayesian hypergraphs that maximally exploits unconditional information while respecting theoretical constraints. For PID-based causal discovery, we establish that direct causal neighbors correspond to variables providing unique information, while co-parents exhibit distinctive synergistic patterns. These results enable a “localist” approach where researchers can analyze specific variables and their immediate causal neighborhoods without reconstructing entire networks. We extend this framework to Bayesian hypergraphs to accommodate more complex causal relationships. Our contributions bridge information theory and graphical causal models, providing both theoretical insights into fundamental limits of observational causal discovery and practical procedures for extracting causal information from data.

Declaration of Originality

I hereby declare that the work presented in this thesis is my own unless otherwise stated. To the best of my knowledge the work is original and ideas developed in collaboration with others have been appropriately referenced.

Acknowledgments

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Introduction

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1.1 Motivation

Understanding higher-order interactions and causal relationships from observational data represents one of the fundamental challenges in modern data science. While traditional statistical methods excel at capturing pairwise correlations, many real-world phenomena emerge from complex interactions among multiple variables that cannot be adequately described by simple correlation structures. In fields ranging from neuroscience and genomics to economics and social sciences, the ability to identify genuine causal mechanisms and distinguish them from spurious associations is crucial for making reliable predictions, designing effective interventions, and building robust scientific theories. The development of principled methods that can uncover these complex interaction patterns while providing causal insights has profound implications for advancing our understanding of complex systems.

This thesis pursues two complementary research goals that address fundamental questions in the intersection of information theory and causal inference.

Exploring the Limits and Power of Unconditional Tests for Causal Discovery. Traditional approaches to causal discovery, particularly those based on Bayesian networks, rely heavily on conditional independence tests to identify Markov equivalence classes of causal structures. While these conditional tests are theoretically powerful and can distinguish between many different causal configurations, they come with significant computational costs that grow combinatorially with the number of conditioning variables. In contrast, interaction measures based on unconditional tests offer a computationally tractable alternative, requiring only pairwise and marginal statistics that can be efficiently estimated from data.

However, unconditional tests are inherently less powerful than their conditional counterparts—there exist causal properties that conditional independence tests can detect but unconditional tests cannot. This raises fundamental questions: What exactly are the limits of unconditional tests for

causal discovery? And how can we leverage the computational advantages of unconditional tests to their fullest extent within these theoretical constraints?

A key feature that emerges when working with unconditional tests is that they often reveal statistical dependencies between variables without providing clear information about the causal direction of these relationships. Traditional directed graphical models like Bayesian networks may be inadequate to represent the partial causal information that unconditional tests can reliably extract. To address this limitation, we turn to Bayesian hypergraphs, which combine the directed relationships of Bayesian networks with the undirected relationships of Markov random fields, providing a more flexible framework for representing the mixed directional and non-directional causal information obtainable from unconditional tests.

Investigating the Relationship Between Partial Information Decomposition and Causality. Partial Information Decomposition (PID) provides a principled framework for decomposing the mutual information between a target variable and a set of source variables into interpretable components: unique information (information about the target that can only be derived from a single source variable), redundant information (information that is shared across multiple source variables), and synergistic information (information that emerges only when multiple source variables are considered together).

A fundamental insight motivating this research direction is that PID components should, in principle, allow us to determine all conditional independence properties of a system, which are the primary tools for causal discovery. This connection arises because having access to all PID components enables us to compute all conditional mutual information terms, and conditional independence is mathematically equivalent to vanishing conditional mutual information. This suggests a deep theoretical link between information decomposition and causal structure.

Given this starting point, we investigate two complementary questions: First, what are the characteristic PID signatures of different causal roles that variables can play? For instance, if variable X is a causal co-parent of variable Y , what can we predict about the PID components involving Y when X serves as the target variable? Second, we explore the converse relationship: if the PID components satisfy particular mathematical conditions, does this imply specific types of causal relationships between the variables under suitable assumptions such as faithfulness and persistent relevance? These questions aim to establish a bidirectional correspondence between information-theoretic quantities and causal structural properties, potentially opening new avenues for causal discovery based on information decomposition that complements conditional independence testing.

A primary motivation for leveraging partial information decomposition for causal discovery is to enable a “localist interpretation” of causal roles that complements traditional global approaches. The central framework of causal discovery typically relies on systematic conditional independence queries across the joint distribution to eliminate possible graphical models and thereby identify causal relationships between variables. This represents a fundamentally globalist approach that operates on the entire joint distribution and considers the complete space of possible graphical models simultaneously.

In contrast, a localist approach allows us to focus on a particular variable of interest and analyze its immediate causal neighborhood in detail. This perspective is especially valuable in situations involving complex interactions among many variables, where we may only care about a subset of variables of primary interest and the specific factors that could be influencing them. Rather than attempting to reconstruct the entire causal network, we can concentrate our analysis on understanding how information flows to and from our target variables.

The PID framework, with its explicit decomposition of informational influences on a designated target variable, is naturally aligned with this localist agenda. By systematically analyzing how different source variables contribute unique, redundant, and synergistic information to a target of interest, we can characterize the local causal environment without requiring complete knowledge of the global causal structure. This approach may offer both computational advantages and interpretive clarity for practitioners interested in understanding specific causal relationships rather

than comprehensive network reconstruction.

1.2 Contributions

Given our two primary motivations for this thesis, we have made the following contributions with respect to our objectives of utilizing unconditional tests and PID for causal discovery.

Unconditional Tests for Causal Discovery. We have provided an original proof for a known technique that utilizes the Lancaster interaction measure to detect V-structures, establishing the theoretical foundation for this approach. Building on this foundation, we systematically identified all three-variable causal structures that are identifiable through unconditional independence tests, while simultaneously characterizing their fundamental limitations for causal discovery.

Motivated by these limitations, we introduced the formal notion of unconditional independence equivalence classes, which precisely captures the causal roles that can and cannot be distinguished using unconditional tests alone. This theoretical framework provides a crisp mathematical characterization of the boundaries of unconditional causal inference.

Finally, we developed a comprehensive procedure that leverages Bayesian hypergraphs to identify all causal structures detectable by unconditional tests. This procedure maximally exploits the power of unconditional tests while respecting the theoretical limits we have established, providing a principled approach to unconditional causal discovery.

PID for Causal Discovery. We proved that under appropriate distributional assumptions, direct causal neighbors of variable X correspond exactly to variables that provide unique information about X . Complementing this result, we demonstrated that co-parents of X are characterized by having zero unique information about X while exhibiting synergistic information with respect to their joint children.

The combination of these two PID signatures provides a systematic recipe for analyzing the immediate causal neighborhood of any variable: the unique information criterion enables identification of direct causal neighbors, while the co-parent synergy pattern allows us to distinguish between child and parent relationships among these neighbors.

We further extended this framework by deriving the characteristic PID signatures of hyperedges in Bayesian hypergraphs, thereby extending our approach to represent the more fine-grained causal relationships expressible through hypergraph structures. This generalization enables the application of information-theoretic causal discovery to more complex dependency patterns that cannot be adequately captured by traditional pairwise graphical models.

1.3 Thesis Structure

In the background chapter, we establish the theoretical foundations necessary for our subsequent analysis. We first introduce unconditional interaction measures, particularly the Lancaster interaction measure and its lattice-theoretic formulation, which serve as the basis for our later results on leveraging unconditional tests for causal discovery. We then provide a comprehensive treatment of graphical models, including Bayesian networks for directed relationships, Markov random fields for undirected relationships, and Bayesian hypergraphs for representing mixtures of directed and undirected interactions. Finally, we introduce the partial information decomposition framework and establish the theoretical connection between PID components and conditional independence structure that motivates our exploration of using PID for causal discovery.

The results section presents our findings across both research objectives, with detailed contributions outlined in Section 1.2. For unconditional tests and causal discovery, we establish theoretical

boundaries through three-variable structure analysis and equivalence class characterization, culminating in our hypergraph-based discovery procedure. For PID type signature of causal roles, we derive the information-theoretic signatures that characterize direct causal relationships and co-parent structures, then extending these patterns to hypergraph representations.

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Background

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2.1 Interaction measures and unconditional independence tests

An interaction measure [1] associated with a multivariate joint probability distribution $P(X_1, \dots, X_D)$ over random variables $(X_1, \dots, X_D)$ is a signed measure ΔP that vanishes whenever P can be factorized in a non-trivial way as a product of its (possibly multivariate) marginal distribution.

Definition 2.1.1. The Lancaster interaction measure is defined by

$$\Delta_L P = \prod_{i=1}^D (P_{X_i}^* - P_{X_i}), \quad (2.1)$$

where each product $\prod_{j=1}^{D'} P_{X_{i_j}}^*$ signifies the joint probability distribution $P_{X_{i_1}, \dots, X_{i_{D'}}}$ of a subvector $(X_{i_1}, \dots, X_{i_{D'}})$.

For a bivariate distribution, the Lancaster interaction measure is simply the difference between the joint probability distribution and the product of the marginal distributions:

$$\Delta_L P = P_{XY} - P_X P_Y, \quad (2.2)$$

Clearly, for $D = 2$ we have $\Delta_L P = 0 \iff x \perp y | \emptyset$

For the case $D = 3$, we obtain

$$\Delta_L P = P_{XYZ} - P_{XY}P_Z - P_{YZ}P_X - P_{XZ}P_Y + 2P_X P_Y P_Z. \quad (2.3)$$

Remark 2.1.1 (Three-Variable Independence Example). To illustrate how the Lancaster interaction measure detects independence, consider three random variables X , Y , and Z . Suppose that $P_{XYZ} = P_X P_{YZ}$, meaning the joint distribution factorizes with X independent of (Y, Z) .

Then by marginalization over Y , we have $P_{XZ} = P_X P_Z$. Hence:

$$P_{XZ}P_Y = P_X P_Y P_Z \quad (2.4)$$

Using a similar argument we can also show that $P_{XY}P_Z = P_X P_Y P_Z$.

Substituting into the Lancaster measure:

$$\Delta_L P = P_{XYZ} - P_{XY}P_Z - P_{YZ}P_X - P_{XZ}P_Y + 2P_X P_Y P_Z \quad (2.5)$$

$$= P_X P_Y P_Z - P_X P_Y P_Z - P_X P_Y P_Z - P_X P_Y P_Z + 2P_X P_Y P_Z = 0 \quad (2.6)$$

This demonstrates that any factorization of the joint distribution into marginals results in a vanishing Lancaster interaction measure, confirming the definition's goal of detecting statistical independence.

Definition 2.1.2 (Lattice). A lattice [2] is a partially ordered set $(\mathcal{L}, \leq)$ defined on a set $\mathcal{X}$ with the property that every pair of elements $x, y \in \mathcal{L}$ possesses both a unique least upper bound, called the *supremum* or *join* (denoted $x \vee y$), and a unique greatest lower bound, called the *infimum* or *meet* (denoted $x \wedge y$).

The lattice structure includes distinguished elements: the maximum element $\mathbf{1}$ (top element) and the minimum element $\mathbf{0}$ (bottom element). A subset of $\mathcal{L}$ that remains closed under both the join and meet operations constitutes a sublattice of $\mathcal{L}$.

Definition 2.1.3 (Möbius Inversion Formula). Let $(\mathcal{L}, \leq)$ be a locally finite partially ordered set (poset) and let $f, g : \mathcal{L} \rightarrow \mathbb{R}$ be functions defined on the lattice. The *Möbius inversion formula* [2] establishes the relationship:

If

$$g(x) = \sum_{y \leq x} f(y), \quad (2.7)$$

then

$$f(x) = \sum_{y \leq x} \mu(y, x) g(y), \quad (2.8)$$

where $\mu(y, x)$ is the *Möbius function* defined recursively by:

$$\mu(x, y) = 1 \quad \text{if } x = y, \quad (2.9)$$

$$\mu(x, y) = - \sum_{z: x \leq z < y} \mu(x, z) \quad \text{for } x < y, \quad (2.10)$$

$$\mu(x, y) = 0 \quad \text{otherwise} \quad (2.11)$$

In the discrete case where the partial order is encoded by the Zeta matrix ζ with elements $\zeta(x, y) = 1$ if $x \leq y$ and 0 otherwise, the Möbius matrix M is the inverse of the Zeta matrix: $M = \zeta^{-1}$.

Remark 2.1.2 (Cancellation Property of Möbius Inversion). The key insight behind Möbius inversion is that the Möbius matrix coefficients are strategically arranged with positive and negative values to ensure systematic cancellation of lower-order terms, isolating the desired “pure” contribution at each lattice element.

Example 2.1.1 (Möbius Inversion on Chain Lattice $\mathbb{Z}_3$). Consider the chain lattice $\mathbb{Z}_3 = \{0, 1, 2\}$

with the natural order $0 \leq 1 \leq 2$. The cumulative sum function gives us:

$$g(0) = f(0) \quad (2.12)$$

$$g(1) = f(0) + f(1) \quad (2.13)$$

$$g(2) = f(0) + f(1) + f(2) \quad (2.14)$$

The Zeta matrix is: $\zeta = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$,

and its inverse, the Möbius matrix, is: $M = \zeta^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$.

Applying Möbius inversion to recover $f(2)$:

$$f(2) = \sum_{y \leq 2} \mu(y, 2)g(y) = \mu(0, 2)g(0) + \mu(1, 2)g(1) + \mu(2, 2)g(2) \quad (2.15)$$

$$= 0 \cdot g(0) + (-1) \cdot g(1) + 1 \cdot g(2) = -g(1) + g(2) \quad (2.16)$$

$$= -(f(0) + f(1)) + (f(0) + f(1) + f(2)) = f(2) \quad (2.17)$$

Note how the negative coefficient -1 applied to $g(1)$ cancels out the unwanted terms $f(0) + f(1)$, leaving only the pure contribution $f(2)$.

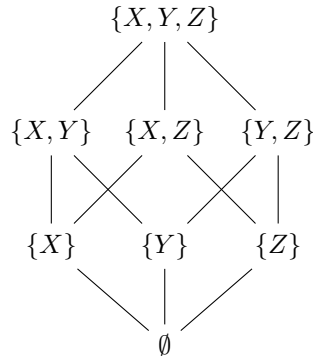
Definition 2.1.4 (Subset Lattice). The *subset lattice* (or Boolean lattice) [2] on a set $\mathcal{X} = \{X_1, X_2, \dots, X_n\}$ is the lattice $(\mathcal{P}(\mathcal{X}), \subseteq)$, where $\mathcal{P}(\mathcal{X})$ denotes the power set of $\mathcal{X}$ and the partial order is given by set inclusion $\subseteq$.

For subsets $A, B \subseteq \mathcal{X}$:

- The join operation is $A \vee B = A \cup B$ (set union)
- The meet operation is $A \wedge B = A \cap B$ (set intersection)
- The bottom element is $\mathbf{0} = \emptyset$ (empty set)
- The top element is $\mathbf{1} = \mathcal{X}$ (full set)

Each subset $S \subseteq \mathcal{X}$ corresponds to a factorization of the joint probability distribution as $P_S P_{S^c}$, where $S^c = \mathcal{X} \setminus S$ is the complement of S .

Example 2.1.2 (Subset Lattice for Three Variables). For $\mathcal{X} = \{X, Y, Z\}$, the subset lattice $(\mathcal{P}(\mathcal{X}), \subseteq)$ contains $2^3 = 8$ elements:



Definition 2.1.5 (Partition). A *partition* [2] of a set $\mathcal{X} = \{X_1, X_2, \dots, X_n\}$ is a collection $\pi = \{B_1, B_2, \dots, B_k\}$ of non-empty, pairwise disjoint subsets (called *blocks*) such that $\bigcup_{i=1}^k B_i = \mathcal{X}$ and $i \neq j \implies B_i \cap B_j = \emptyset$. We write $\pi \vdash \mathcal{X}$ to denote that π is a partition of $\mathcal{X}$.

Two elements $x, y \in \mathcal{X}$ are said to be *equivalent* under partition π , denoted $x \sim_\pi y$, if they belong to the same block of π .

Definition 2.1.6 (Refinement Relation). Let π and σ be two partitions of a set $\mathcal{X}$. We say that π *refines* σ , denoted $\pi \preceq \sigma$ (or “ π is finer than σ ”), if every block of π is contained within some block of σ .

Equivalently, $\pi \preceq \sigma$ if and only if for all $x, y \in \mathcal{X}$: if $x \sim_\pi y$, then $x \sim_\sigma y$. That is, if two elements are equivalent under π , they must also be equivalent under σ .

When $\pi \preceq \sigma$, we say σ is *coarser* than π .

Definition 2.1.7 (Partition Lattice). The *partition lattice* (or lattice of set partitions) [2] on a set $\mathcal{X}$ is the lattice $(\Pi(\mathcal{X}), \preceq)$, where $\Pi(\mathcal{X})$ denotes the set of all partitions of $\mathcal{X}$ and the partial order $\preceq$ is defined by refinement.

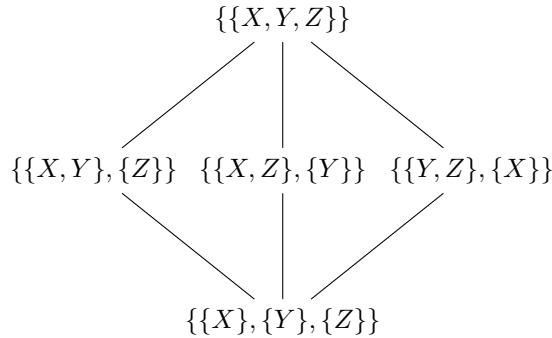
For partitions $\pi, \sigma \in \Pi(\mathcal{X})$, we say $\pi \preceq \sigma$ (read “ π refines σ ” or “ π is finer than σ ”) if every block of π is contained within some block of σ . Equivalently, $\pi \preceq \sigma$ if and only if $x \sim_\pi y$ implies $x \sim_\sigma y$ for all $x, y \in \mathcal{X}$.

The lattice operations are:

- The join $\pi \vee \sigma$ is the *coarsest* partition that is finer than both π and σ
- The meet $\pi \wedge \sigma$ is the *finest* partition that is coarser than both π and σ
- The bottom element is $\mathbf{0} = \{\{x\} : x \in \mathcal{X}\}$ (discrete partition)
- The top element is $\mathbf{1} = \{\mathcal{X}\}$ (trivial partition)

Each partition $\pi = \{B_1, B_2, \dots, B_k\} \in \Pi(\mathcal{X})$ corresponds to a factorization of the joint probability distribution: $P_\pi = \prod_{i=1}^k P_{B_i}$, where P_{B_i} is the marginal distribution over variables in block B_i .

Example 2.1.3 (Partition Lattice for Three Variables). For $\mathcal{X} = \{X, Y, Z\}$, the partition lattice $(\Pi(\mathcal{X}), \preceq)$ contains 5 elements (Bell number $B_3 = 5$):



The correspondence with probability factorizations is:

$$\{\{X\}, \{Y\}, \{Z\}\} \leftrightarrow P_X P_Y P_Z \quad (2.18)$$

$$\{\{X, Y\}, \{Z\}\} \leftrightarrow P_{XY} P_Z \quad (2.19)$$

$$\{\{X, Z\}, \{Y\}\} \leftrightarrow P_{XZ} P_Y \quad (2.20)$$

$$\{\{Y, Z\}, \{X\}\} \leftrightarrow P_{YZ} P_X \quad (2.21)$$

$$\{\{X, Y, Z\}\} \leftrightarrow P_{XYZ} \quad (2.22)$$

Definition 2.1.8 (Kullback-Leibler Divergence). Let P and Q be two probability distributions over the same sample space $\mathcal{X}$. The *Kullback-Leibler (KL) divergence* [2] from Q to P is defined as:

$$D_{\text{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)} \quad (2.23)$$

for discrete distributions, or

$$D_{\text{KL}}(P \parallel Q) = \int_{\mathcal{X}} p(x) \log \frac{p(x)}{q(x)} dx \quad (2.24)$$

for continuous distributions with densities $p(x)$ and $q(x)$.

The KL divergence is always non-negative, $D_{\text{KL}}(P \parallel Q) \geq 0$, with equality if and only if $P = Q$ almost everywhere.

Remark 2.1.3 (Intuition: KL Divergence as Excess Surprise). To understand the meaning of KL divergence, we first recall that the *entropy* of a probability distribution P is defined as:

$$H(P) = - \sum_{x \in \mathcal{X}} P(x) \log P(x) \quad (2.25)$$

Entropy measures the average amount of *surprise* or *uncertainty* inherent in the distribution P . When an event x occurs with probability $P(x)$, the “surprise” or “information content” of observing x is $-\log P(x)$ (low probability events are more surprising). The entropy is thus the expected surprise under distribution P .

The KL divergence $D_{\text{KL}}(P \parallel Q)$ can be interpreted as measuring the *excess surprise* we experience when we incorrectly assume the distribution is Q but the true distribution is actually P . Specifically:

- If we assume distribution Q , we expect the surprise of event x to be $-\log Q(x)$
- However, if the true distribution is P , event x actually occurs with frequency $P(x)$
- The average surprise we experience is $\sum_x P(x) \cdot (-\log Q(x)) = -\sum_x P(x) \log Q(x)$, called the *cross-entropy* $H(P, Q)$
- The entropy $H(P) = -\sum_x P(x) \log P(x)$ represents the minimum possible average surprise (achieved when our assumed distribution matches the true one)

Therefore, the KL divergence measures how much *additional surprise* we incur by using the wrong distribution:

$$D_{\text{KL}}(P \parallel Q) = H(P, Q) - H(P) = (\text{surprise with wrong model}) - (\text{optimal surprise}) \quad (2.26)$$

This excess surprise quantifies how “different” the distributions P and Q are: the more they differ, the more extra surprise we experience by using Q instead of P .

Remark 2.1.4 (KL Divergence as Independence Measure). The choice of $g(\cdot) = D_{\text{KL}}(\cdot \parallel P_1 P_2 \cdots P_d)$ as the sum function on the lattice provides an elegant way to measure statistical independence using the intuition that KL divergence quantifies distributional “difference.”

Recall that random variables $X_1, \dots, X_d$ are *statistically independent* if and only if their joint distribution factorizes as the product of marginals:

$$P_{X_1, \dots, X_d} = P_{X_1} P_{X_2} \cdots P_{X_d} \quad (2.27)$$

Since KL divergence measures how much two probability distributions differ, we have:

$$D_{\text{KL}}(P_{X_1, \dots, X_d} \parallel P_{X_1} P_{X_2} \cdots P_{X_d}) = 0 \iff \text{complete independence} \quad (2.28)$$

More generally, for any partition $\pi \in \Pi(\{X_1, \dots, X_d\})$, the KL divergence $D_{\text{KL}}(P_{\{X_1, \dots, X_d\}} \parallel \prod_{B \in \pi} P_B)$ measures how much the joint distribution deviates from the factorization implied by partition π .

Remark 2.1.5 (Interaction Measures via Möbius Inversion of KL Divergence). The goal of an interaction measure is to capture the *unique contribution* that cannot be decomposed into lower-order effects. We can achieve this by combining the lattice structure with KL divergence and Möbius inversion.

Consider the sum function $g(\pi) = D_{\text{KL}}(\prod_{B \in \pi} P_B \parallel P_1 P_2 \cdots P_d)$ on the partition lattice, which measures the “excess surprise” when we assume total independence $P_1 P_2 \cdots P_d$ but the true distribution is the factorization $\prod_{B \in \pi} P_B$ corresponding to partition π .

The key insight is that if $g(\pi_1) = g(\pi_2)$ for two different partitions π_1 and π_2 , then partition π_1 provides *no unique contribution* beyond what π_2 already captures. For example:

- If $g(\{\{X, Y, Z\}\}) = g(\{\{X\}, \{Y, Z\}\})$, then the three-way dependence P_{XYZ} contributes no additional surprise beyond the two-way factorization $P_X P_{YZ}$
- This means $D_{\text{KL}}(P_{XYZ} \parallel P_X P_Y P_Z) = D_{\text{KL}}(P_X P_{YZ} \parallel P_X P_Y P_Z)$, indicating that the deviation from independence is fully captured by the YZ dependence

The Möbius inversion formula systematically isolates these unique contributions:

$$f(\pi) = \sum_{\sigma \preceq \pi} \mu(\sigma, \pi) g(\sigma) \quad (2.29)$$

Here, $f(\pi)$ represents the *irreducible interaction* specific to partition π that cannot be explained by any finer partition. The Möbius coefficients $\mu(\sigma, \pi)$ ensure that:

- When $g(\pi)$ can be “explained” by finer partitions, the contributions cancel out and $f(\pi) = 0$
- When $g(\pi)$ contains genuine new structure beyond its refinements, $f(\pi) \neq 0$ captures this unique effect

This is exactly what we want from an interaction measure: *the interaction will be zero whenever any non-trivial factorization is satisfied*. If the joint probability distribution P_{XYZ} can be factorized as $P_X P_{YZ}$ (or any other non-trivial factorization), then there is no unique contribution from the full joint distribution that isn’t already captured by that simpler factorization. The interaction measure should detect this and return zero, indicating no irreducible higher-order dependence.

This transforms our KL divergence measurements of “excess surprise relative to total independence” into a hierarchical decomposition where each $f(\pi)$ isolates the pure interaction effects that emerge precisely at the level of partition π .

2.2 Graphical models and Causality

While interaction measures provide valuable tools for detecting non-trivial factorizations of joint probability distributions using unconditional independence tests, they primarily focus on whether a distribution can be decomposed into products of (possibly multivariate) marginal distributions.

To achieve more fine-grained understanding of causal relationships, we require more sophisticated probabilistic models that can capture the detailed dependency structure among variables.

Graphical models offer precisely this capability by enabling fine-grained factorizations of probability distributions according to underlying graph structures. Bayesian networks, for instance, factorize the joint probability distribution as products of various conditional probability terms specified by the directed acyclic graph, with each variable depending only on its immediate parents. This factorization structure implies specific conditional independence properties that can be systematically analyzed using principles such as d-separation. Undirected graphical models, on the other hand, allow us to represent relationships such as "variables A and B are related to each other, but only through variable C ," ensuring that once we condition on C , variables A and B become conditionally independent.

The general framework of causal discovery operates on the principle that, given a joint probability distribution over all variables of interest, we can perform various conditional independence queries to systematically probe the dependency structure. The pattern of conditional independence relationships that we observe allows us to narrow down the space of possible causal structures to an equivalence class of graphical models that are consistent with these statistical properties. This approach transforms the problem of causal discovery from one of pure statistical association into a principled inference procedure based on the conditional independence structure encoded in probability distributions.

2.2.1 Bayesian networks

Definition 2.2.1 (Directed Graph). A *directed graph* [3, Chapter 8] is a pair $G = (V, E)$ where:

- V is a finite set of vertices (or nodes)
- $E \subseteq V \times V$ is a set of directed edges (or arcs)

A directed edge from vertex u to vertex v is denoted as $(u, v) \in E$ or $u \rightarrow v$, indicating a direction from u to v . Unlike undirected graphs, the edge (u, v) is distinct from (v, u) .

Definition 2.2.2 (Cycle in Directed Graph). A *cycle* in a directed graph [3, Chapter 8] is a sequence of vertices $v_1, v_2, \dots, v_k, v_1$ where:

- $k \geq 2$
- All vertices $v_1, v_2, \dots, v_k$ are distinct
- There exists a directed edge (v_i, v_{i+1}) for each $i = 1, 2, \dots, k - 1$
- There exists a directed edge (v_k, v_1) completing the cycle

In other words, a cycle is a directed path that starts and ends at the same vertex, visiting each intermediate vertex exactly once.

Definition 2.2.3 (Directed Acyclic Graph). A *directed acyclic graph* (DAG) [3, Chapter 8] is a directed graph $G = (V, E)$ that contains no cycles. Equivalently, a DAG is a directed graph in which there is no sequence of directed edges that forms a closed loop.

An important property of DAGs is that they admit a *topological ordering*: the vertices can be arranged in a linear order such that for every directed edge (u, v) , vertex u appears before vertex v in the ordering.

Definition 2.2.4 (Parents, Children, Ancestors, and Descendants in DAGs). Given a directed acyclic graph $G = (V, E)$ and a vertex $v \in V$ [3, Chapter 8]:

- The *parents* of v , denoted $\text{pa}(v)$, are the set of vertices with directed edges to v : $\text{pa}(v) = \{u \in V : (u, v) \in E\}$
- The *children* of v , denoted $\text{ch}(v)$, are the set of vertices with directed edges from v : $\text{ch}(v) = \{u \in V : (v, u) \in E\}$
- The *ancestors* of v , denoted $\text{an}(v)$, are all vertices from which there exists a directed path to v
- The *descendants* of v , denoted $\text{de}(v)$, are all vertices to which there exists a directed path from v

Definition 2.2.5 (Bayesian Network Factorization). A *Bayesian network* [3, Chapter 8] is a directed acyclic graph $G = (V, E)$ together with a joint probability distribution that factorizes according to the graph structure. Specifically, if $X_1, X_2, \dots, X_n$ are random variables corresponding to vertices $v_1, v_2, \dots, v_n$, then the joint distribution factorizes as:

$$p(x_1, x_2, \dots, x_n) = \prod_{i=1}^n p(x_i \mid \text{pa}(x_i)) \quad (2.30)$$

where $\text{pa}(x_i)$ denotes the parents of variable X_i in the graph, and we use the convention that $p(x_i \mid \emptyset) = p(x_i)$ when X_i has no parents.

Definition 2.2.6 (Conditional Independence). Two random variables X and Y are *conditionally independent* given a third random variable Z [3, Chapter 8], denoted $X \perp Y \mid Z$, if:

$$p(x, y \mid z) = p(x \mid z)p(y \mid z) \quad (2.31)$$

for all values x , y , and z where the distributions are defined.

Equivalently, conditional independence can be expressed as:

$$p(x \mid y, z) = p(x \mid z) \quad (2.32)$$

or

$$p(y \mid x, z) = p(y \mid z) \quad (2.33)$$

When $Z = \emptyset$ (the empty set), conditional independence reduces to marginal independence: $X \perp Y$, meaning $p(x, y) = p(x)p(y)$.

Example 2.2.1 (Chain Structure Factorization). Consider the chain structure $A \rightarrow C \rightarrow B$ where:

- A has no parents: $\text{pa}(A) = \emptyset$
- C has parent A : $\text{pa}(C) = \{A\}$
- B has parent C : $\text{pa}(B) = \{C\}$

The Bayesian network factorization gives:

$$p(a, c, b) = p(a) \cdot p(c \mid a) \cdot p(b \mid c) \quad (2.34)$$

We can verify that conditioning on the parent C renders the ancestor A irrelevant for B . To show $p(b \mid c) = p(b \mid a, c)$:

$$p(b \mid a, c) = \frac{p(a, b, c)}{p(a, c)} \quad (2.35)$$

$$= \frac{p(a) \cdot p(c \mid a) \cdot p(b \mid c)}{p(a, c)} \quad (\text{by factorization}) \quad (2.36)$$

$$= \frac{p(a) \cdot p(c \mid a) \cdot p(b \mid c)}{p(a) \cdot p(c \mid a)} \quad (\text{since } p(a, c) = p(a)p(c \mid a)) \quad (2.37)$$

$$= p(b \mid c) \quad (2.38)$$

This proves that once we condition on the parent C , conditioning on the ancestor A provides no additional information about B . In other words, A and B are conditionally independent given C : $B \perp A \mid C$.

Remark 2.2.1 (Ruling Out Ancestral Influence). The key insight of Bayesian networks is that they allow us to “rule out influence” from distant ancestors [3, Chapter 8]. In the factorization $p(x_i \mid \text{pa}(x_i))$, each variable depends only on its immediate parents, not on the entire set of ancestors.

This conditional independence structure means that once we condition on a variable’s parents, its ancestors become irrelevant for prediction. For instance, in our chain example, $p(b \mid ca) = p(b \mid c)$ because the influence of A on B is fully mediated through C .

This property enables computational efficiency by avoiding marginalization over large numbers of variables. Instead of computing complex marginal distributions involving many ancestors, we only need to consider immediate parent-child relationships, dramatically reducing the dimensionality of probabilistic inference problems.

Definition 2.2.7 (Three-Variable Graphical Structures). For three random variables A , B , and C , there are three fundamental graphical structures that form the basis for understanding conditional independence in Bayesian networks [3, Chapter 8]:

1. **Chain structure:** $A \rightarrow C \rightarrow B$
2. **Confounder structure:** $A \leftarrow C \rightarrow B$
3. **V-structure (Collider):** $A \rightarrow C \leftarrow B$

Example 2.2.2 (Chain Structure: $A \rightarrow C \rightarrow B$). Factorization: $p(a, c, b) = p(a)p(c \mid a)p(b \mid c)$

Conditional independence property: $A \perp B \mid C$ but $A \not\perp B \mid \emptyset$

As proven earlier, conditioning on C renders A and B conditionally independent.

Example 2.2.3 (Confounder Structure: $A \leftarrow C \rightarrow B$). Factorization: $p(a, c, b) = p(c)p(a \mid c)p(b \mid c)$

Conditional independence property: $A \perp B \mid C$

Proof: We show that $p(a \mid b, c) = p(a \mid c)$:

$$p(a \mid b, c) = \frac{p(a, b, c)}{p(b, c)} \quad (2.39)$$

$$= \frac{p(c)p(a \mid c)p(b \mid c)}{p(b, c)} \quad (\text{by factorization}) \quad (2.40)$$

$$= \frac{p(c)p(a \mid c)p(b \mid c)}{p(c)p(b \mid c)} \quad (\text{since } p(b, c) = p(c)p(b \mid c)) \quad (2.41)$$

$$= p(a \mid c) \quad (2.42)$$

Therefore, $A \perp B \mid C$. However, $A \not\perp B$ marginally, since both depend on the common cause C .

Example 2.2.4 (V-structure (Collider): $A \rightarrow C \leftarrow B$). Factorization: $p(a, c, b) = p(a)p(b)p(c \mid a, b)$

Conditional independence properties:

- $A \perp B$ (marginally independent)
- $A \not\perp B \mid C$ (conditionally dependent given C)

Proof of marginal independence:

$$p(a, b) = \sum_c p(a, b, c) \quad (2.43)$$

$$= \sum_c p(a)p(b)p(c \mid a, b) \quad (2.44)$$

$$= p(a)p(b) \sum_c p(c \mid a, b) = p(a)p(b) \quad (2.45)$$

Proof that conditioning creates dependence: By Bayes' theorem [3, Chapter 8]:

$$p(a \mid b, c) = \frac{p(a, b, c)}{p(b, c)} = \frac{p(a)p(b)p(c \mid a, b)}{p(b, c)} \quad (2.46)$$

$$= \frac{p(a)p(b)p(c \mid a, b)}{\sum_{a'} p(a', b, c)} = \frac{p(a)p(b)p(c \mid a, b)}{\sum_{a'} p(a')p(b)p(c \mid a', b)} \quad (2.47)$$

$$= \frac{p(a)p(c \mid a, b)}{\sum_{a'} p(a')p(c \mid a', b)} \quad (2.48)$$

Since $p(c \mid a, b)$ generally depends on both a and b , we have $p(a \mid b, c) \neq p(a \mid c)$, so $A \not\perp B \mid C$.

This is the *collider effect*: conditioning on a common effect C creates dependence between its causes A and B , even though they are marginally independent.

Remark 2.2.2 (Summary of Conditional Independence Properties). The three fundamental structures exhibit different conditional independence behaviors [3, Chapter 8]:

- **Chain** ($A \rightarrow C \rightarrow B$): $A \perp B \mid C$ but $A \not\perp B \mid \emptyset$ (conditioning blocks the path)
- **Confounder** ($A \leftarrow C \rightarrow B$): $A \perp B \mid C$ but $A \not\perp B \mid \emptyset$ (conditioning blocks the common cause)
- **Collider** ($A \rightarrow C \leftarrow B$): $A \perp B$ but $A \not\perp B \mid C$ (conditioning creates dependence)

These patterns form the foundation for d-separation, which determines conditional independence in arbitrary directed acyclic graphs.

Remark 2.2.3 (Intuition: Information Flow in Graphs). We can think of (conditional) dependence as information being able to “flow” from one variable to another variable through the graph structure [3, Chapter 8]. This flow metaphor provides intuitive understanding of d-separation:

- **Multiple paths**: If there are multiple paths between two variables, we only need *one* valid (unblocked) path for information to flow, establishing dependence
- **Path segments**: For any particular path to be valid, *every* segment along that path must be unblocked/open

- **Segment analysis:** Each segment corresponds to one of the three fundamental structures (chain, confounder, or collider), whose conditional independence properties we can determine using our earlier analysis

The d-separation criterion formalizes this intuition by providing rules for when paths are “blocked” or “open” given a conditioning set.

Definition 2.2.8 (Undirected Path). An *undirected path* in a directed acyclic graph [3, Chapter 8] between vertices A and B is a sequence of vertices $A = V_0, V_1, V_2, \dots, V_n = B$ such that for each consecutive pair (V_{i-1}, V_i) , there exists either a directed edge $V_{i-1} \rightarrow V_i$ or $V_i \rightarrow V_{i-1}$ in the graph.

Note that an undirected path ignores the direction of edges—we can traverse edges in either direction when following the path.

Definition 2.2.9 (d-Separation). Let A , B , and S be three disjoint sets of vertices in a directed acyclic graph G . The sets A and B are *d-separated* by S , denoted by $A \perp_d B \mid S$, if every undirected path between any vertex in A and any vertex in B is *blocked* by S [3, Chapter 8].

An undirected path is **blocked** by conditioning set S if there exists at least one vertex V on the path such that one of the following conditions holds:

1. **Chain structure:** V is part of a chain $\dots \rightarrow V \rightarrow \dots$ or $\dots \leftarrow V \leftarrow \dots$, and $V \in S$
2. **Confounder structure:** V is part of a confounder $\dots \leftarrow V \rightarrow \dots$, and $V \in S$
3. **Collider structure:** V is part of a collider $\dots \rightarrow V \leftarrow \dots$, and *neither* V nor any descendant of V is in S

Conversely, a path is **open** (unblocked) if *none* of the blocking conditions apply, meaning:

1. For every chain/confounder vertex V on the path: $V \notin S$
2. For every collider vertex V on the path: $V \in S$ or some descendant of V is in S

If A and B are d-separated by S , then A and B are conditionally independent given S in any probability distribution that factorizes according to the graph.

Remark 2.2.4 (Connection to Three-Variable Structure Analysis). The d-separation blocking conditions directly correspond to the conditional independence properties of the three fundamental structures analyzed earlier [3, Chapter 8]:

- **Chain** ($A \rightarrow C \rightarrow B$): We proved that $A \perp B \mid C$, meaning conditioning on C blocks the path. In d-separation terms, when $C \in S$, the chain segment is blocked.
- **Confounder** ($A \leftarrow C \rightarrow B$): We proved that $A \perp B \mid C$, meaning conditioning on the common cause C blocks the path. In d-separation terms, when $C \in S$, the confounder segment is blocked.
- **Collider** ($A \rightarrow C \leftarrow B$): We proved that $A \perp B$ marginally but $A \not\perp B \mid C$. This means:
 - When $C \notin S$, the path is blocked (respecting marginal independence)
 - When $C \in S$, the path becomes open (conditioning creates dependence)

The d-separation criterion generalizes these local three-variable rules to arbitrary graph structures by requiring that *every* segment along a path satisfies the appropriate blocking condition. A single unblocked path is sufficient to induce conditional dependence, reflecting the intuition that information needs only one valid route to flow between variables.

Example 2.2.5 (d-Separation Analysis). Consider the path $X_1 \rightarrow X_2 \rightarrow X_3 \leftarrow X_4 \rightarrow X_5$ and the query: Are X_1 and X_5 independent given $S = \{X_3\}$?

Analyzing each segment:

- $X_1 \rightarrow X_2 \rightarrow X_3$: Chain structure, $X_2 \notin S$ (open)
- $X_2 \rightarrow X_3 \leftarrow X_4$: Collider structure at X_3 , $X_3 \in S$ (open)
- $X_3 \leftarrow X_4 \rightarrow X_5$: Confounder structure, $X_4 \notin S$ (open)

Since all segments are open, the path is open, so $X_1 \not\perp X_5 \mid X_3$. The collider X_3 being conditioned upon “activates” the path, allowing information to flow from X_1 to X_5 .

Theorem 2.2.1 (Soundness and Completeness of d-Separation). *Let G be a directed acyclic graph and let P be a probability distribution that factorizes according to G [3, Chapter 8].*

Soundness: *If sets of variables A and B are d-separated by set C in graph G , then A and B are conditionally independent given C in any probability distribution P that factorizes according to G :*

$$A \perp_d B \mid C \text{ in } G \Rightarrow A \perp B \mid C \text{ in } P \quad (2.49)$$

Completeness: *Conversely, if P has strictly positive density and is faithful to G , then any conditional independence relationship that holds in P can be determined by d-separation in G :*

$$A \perp B \mid C \text{ in } P \Rightarrow A \perp_d B \mid C \text{ in } G \quad (2.50)$$

Together, these results establish that for faithful distributions with strictly positive density, d-separation exactly characterizes the conditional independence structure encoded by the directed acyclic graph:

$$A \perp_d B \mid C \text{ in } G \Leftrightarrow A \perp B \mid C \text{ in } P \quad (2.51)$$

Definition 2.2.10 (Faithfulness of Bayesian Networks). A probability distribution P is said to be *faithful* to a directed acyclic graph G [3, Chapter 8] if the conditional independence relationships in P are exactly those that can be determined by d-separation in G .

Formally, P is faithful to G if and only if:

$$X \perp Y \mid Z \text{ in } P \Leftrightarrow X \perp_d Y \mid Z \text{ in } G \quad (2.52)$$

for all disjoint sets of variables X , Y , and Z .

Equivalently, faithfulness means:

- **No missing edges:** If X and Y are conditionally independent given some set Z in the probability distribution, then X and Y must be d-separated by Z in the graph
- **No extra independencies:** The distribution contains no conditional independence relationships beyond those implied by the graph structure

When a distribution is faithful to a graph, the graph provides a complete and accurate representation of the conditional independence structure of the distribution.

Definition 2.2.11 (Graphoid Axioms). Let X , Y , Z , and W be disjoint sets of random variables. A conditional independence relation $\perp$ satisfies the *graphoid axioms* [3, Chapter 8] if the following properties hold:

1. **Symmetry:** $X \perp Y \mid Z \Rightarrow Y \perp X \mid Z$

2. **Decomposition:** $X \perp (Y, Z) \mid W \Rightarrow X \perp Y \mid W$ and $X \perp Z \mid W$
3. **Weak Union:** $X \perp (Y, Z) \mid W \Rightarrow X \perp Y \mid (Z, W)$
4. **Contraction:** $X \perp Y \mid W$ and $X \perp Z \mid (W, Y) \Rightarrow X \perp (Y, Z) \mid W$

When the probability distribution has strictly positive density (i.e., $p(\mathbf{x}) > 0$ for all $\mathbf{x}$), the following additional axiom holds:

5. **Intersection:** $X \perp Y \mid (Z, W)$ and $X \perp Z \mid (Y, W) \Rightarrow X \perp (Y, Z) \mid W$

These axioms capture the intuitive properties that conditional independence relationships should satisfy.

Remark 2.2.5 (Importance of Positive Probability Assumption). The requirement of strictly positive probability density is crucial for the completeness result [3, Chapter 8]. When some configurations have zero probability, additional conditional independence relationships may hold that cannot be derived from the graph structure alone.

For instance, if $p(x_1 = 0, x_2 = 0) = 0$ in a distribution, then X_1 and X_2 might exhibit conditional independence relationships that are not reflected in the graph topology. The positive probability assumption ensures that the graph structure alone determines all conditional independence relationships, without interference from deterministic constraints or boundary effects.

2.2.2 Undirected models and Markov equivalence classes

Bayesian networks are specifically designed to represent causal relationships between variables, capturing the inherent asymmetry that characterizes causal interactions. This asymmetry is fundamental to understanding causal mechanisms: if variable A is a causal parent of variable B , then any factor that correlates with A will necessarily correlate with B through the causal pathway. However, the converse relationship does not hold—correlations with B do not automatically imply correlations with A , since B may have independent causal influences beyond A .

Furthermore, Bayesian networks exhibit the distinctive property that conditioning can introduce new dependencies between variables through the collider effect. When two independent causes converge on a common effect, conditioning on that effect creates statistical dependence between the otherwise independent causes, even though no direct causal relationship exists between them.

In contrast, undirected models represent symmetric dependencies where the statistical relationship from variable A to variable B is fundamentally analogous to that from B to A . These models primarily capture mediation effects, where variables are connected through intermediate pathways. In undirected models, conditioning typically results in greater independence between variables by blocking mediating pathways, rather than creating new dependencies as in the directed case.

Despite these fundamental differences, there exist important connections between Bayesian networks and undirected models. By systematically removing directional information from Bayesian networks—such as through the moralization process—we can obtain coarse-grained undirected representations that preserve certain conditional independence properties while sacrificing the detailed causal asymmetries. This connection allows us to bridge the precise causal information captured by directed models and the computational advantages offered by undirected representations.

Definition 2.2.12 (Undirected Graph). An *undirected graph* [3, Chapter 8] is a pair $G = (V, E)$ where V is a finite set of vertices (or nodes) and $E \subseteq \{\{u, v\} : u, v \in V, u \neq v\}$ is a set of undirected edges. An undirected edge between vertices u and v is denoted as $\{u, v\} \in E$ or $u - v$, indicating a symmetric connection with no inherent direction. Unlike directed graphs, the edge $\{u, v\}$ is identical to $\{v, u\}$.

Definition 2.2.13 (Markov Properties for Undirected Graphs). Let $G = (V, E)$ be an undirected graph with vertices corresponding to random variables. The Markov properties characterize conditional independence relationships encoded by the graph structure [3, Chapter 8]:

Pairwise Markov Property: Two non-adjacent variables are conditionally independent given all other variables:

$$\{X_i, X_j\} \notin E \Rightarrow X_i \perp X_j \mid V \setminus \{X_i, X_j\} \quad (2.53)$$

Local Markov Property: Each variable is conditionally independent of all non-neighbors given its neighbors:

$$X_i \perp (V \setminus \{X_i\} \setminus \text{ne}(X_i)) \mid \text{ne}(X_i) \quad (2.54)$$

where $\text{ne}(X_i) = \{X_j : \{X_i, X_j\} \in E\}$ denotes the neighbors of X_i .

Global Markov Property: Two sets of variables A and B are conditionally independent given a separating set C if every path between A and B passes through C :

$$A \perp B \mid C \quad \text{if } C \text{ separates } A \text{ and } B \text{ in } G \quad (2.55)$$

where C separates A and B if removing vertices in C (and their incident edges) disconnects all paths between vertices in A and vertices in B .

Remark 2.2.6 (Hierarchy of Markov Properties). For undirected graphs, these three Markov properties form a hierarchy [3, Chapter 8]:

$$\text{Global} \Rightarrow \text{Local} \Rightarrow \text{Pairwise} \quad (2.56)$$

The global property is the strongest, implying both local and pairwise properties. The local property implies the pairwise property. For strictly positive distributions, all three properties are equivalent, but for general distributions, the implications may be strict.

Definition 2.2.14 (Clique and Maximal Clique). Let $G = (V, E)$ be an undirected graph [3, Chapter 8]:

A *clique* $C \subseteq V$ is a subset of vertices such that every pair of distinct vertices in C is connected by an edge:

$$\forall X_i, X_j \in C, i \neq j : \{X_i, X_j\} \in E \quad (2.57)$$

A *maximal clique* is a clique that cannot be extended by adding any other vertex while maintaining the clique property. Formally, C is maximal if C is a clique and for any $X_k \notin C$, the set $C \cup \{X_k\}$ is not a clique.

Definition 2.2.15 (Clique Factorization). A probability distribution over an undirected graph $G = (V, E)$ has a *clique factorization* [3, Chapter 8] if it can be written as:

$$p(\mathbf{x}) = \frac{1}{Z} \prod_C \psi_C(\mathbf{x}_C) \quad (2.58)$$

where:

- The product is over all maximal cliques C in the graph
- $\psi_C(\mathbf{x}_C) > 0$ are *potential functions* (or *clique potentials*) defined on the variables $\mathbf{x}_C$ in clique C
- Z is the *partition function* (normalization constant):

$$Z = \sum_{\mathbf{x}} \prod_C \psi_C(\mathbf{x}_C) \quad (2.59)$$

Theorem 2.2.2 (Clique Factorization Satisfies Markov Properties). *If a probability distribution $p(\mathbf{x})$ factorizes according to the cliques of an undirected graph G as in the clique factorization, then it satisfies the pairwise, local, and global Markov properties [3, Chapter 8].*

Definition 2.2.16 (Markov Equivalence Class). Two directed acyclic graphs G_1 and G_2 over the same set of variables V are *Markov equivalent* [3, Chapter 8] if they encode the same set of conditional independence relationships. That is, for any disjoint sets of variables A , B , and C :

$$A \perp_d B \mid C \text{ in } G_1 \Leftrightarrow A \perp_d B \mid C \text{ in } G_2 \quad (2.60)$$

Hence a *Markov equivalence class* is the set of all DAGs that are Markov equivalent to each other. All DAGs within the same Markov equivalence class represent the same statistical model and cannot be distinguished using observational data alone under the faithfulness assumption.

Example 2.2.6 (Converting Bayesian Network to Markov Random Field via Moralization). Consider a Bayesian network with the factorization:

$$p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2)p(x_3)p(x_4 \mid x_1, x_2, x_3) \quad (2.61)$$

The corresponding DAG has x_1 , x_2 , and x_3 as parents of x_4 , with no edges among the parents.

We can interpret each term in the factorization as a potential function over the variables it involves:

$$p(x_1, x_2, x_3, x_4) = p(x_1) \cdot p(x_2) \cdot p(x_3) \cdot p(x_4 \mid x_1, x_2, x_3) \quad (2.62)$$

$$= \psi_1(x_1) \cdot \psi_2(x_2) \cdot \psi_3(x_3) \cdot \psi_{1234}(x_1, x_2, x_3, x_4) \quad (2.63)$$

The crucial observation is that the factor $p(x_4 \mid x_1, x_2, x_3)$ can be viewed as a potential function $\psi_{1234}(x_1, x_2, x_3, x_4)$ that depends on all four variables. For this to be a valid MRF clique potential, all variables $\{x_1, x_2, x_3, x_4\}$ must form a clique in the undirected graph, requiring them to be pairwise connected.

To construct the moral graph, we add edges between all parents of x_4 (creating moral edges $x_1 - x_2$, $x_1 - x_3$, and $x_2 - x_3$) and then drop edge directions. The resulting moral graph is fully connected with edges:

$$\{x_1, x_2\}, \{x_1, x_3\}, \{x_1, x_4\}, \{x_2, x_3\}, \{x_2, x_4\}, \{x_3, x_4\} \quad (2.64)$$

Definition 2.2.17 (Moral Graph). Given a directed acyclic graph $G = (V, E)$, the *moral graph* (or *moralized graph*) G^m [3, Chapter 8] is an undirected graph constructed by the following two-step process:

1. **Add moral edges:** For every vertex $v \in V$, add undirected edges between all pairs of parents of v . That is, if $u, w \in \text{pa}(v)$ and $\{u, w\} \notin E$, then add the undirected edge $\{u, w\}$ to create a “moral edge.”
2. **Drop directions:** Replace all directed edges with undirected edges, yielding an undirected graph $G^m = (V, E^m)$.

The term “moral” refers to the process of “marrying” the parents of each vertex, ensuring that all parents of a common child are connected.

Theorem 2.2.3 (Conditional Independence Preservation in Moralization). *Let $G = (V, E)$ be a directed acyclic graph representing a Bayesian network, and let G^m be its moral graph. If a probability distribution P factorizes according to the Bayesian network G , then P satisfies all conditional independence properties implied by the Markov random field structure of G^m [3, Chapter 8]. Formally, for any disjoint sets of variables A , B , and $C \subseteq V$:*

$$\text{If } C \text{ separates } A \text{ and } B \text{ in } G^m, \text{ then } A \perp B \mid C \text{ in } P. \quad (2.65)$$

In other words, every conditional independence relationship that can be read from the moral graph G^m using the global Markov property for undirected graphs is satisfied by any probability distribution that factorizes according to the original DAG G .

Theorem 2.2.4 (Invariance of Moral Graphs within Markov Equivalence Classes). *Let G_1 and G_2 be two directed acyclic graphs that belong to the same Markov equivalence class. Then their moral graphs are identical [3, Chapter 8]:*

$$G_1^m = G_2^m \quad (2.66)$$

Definition 2.2.18 (Markov Blanket). Given a directed acyclic graph $G = (V, E)$ and a variable $X_i \in V$, the Markov blanket of X_i , denoted by $MB(X_i)$, is the minimal set of variables that renders X_i conditionally independent of all other variables in the graph [3].

Formally, $MB(X_i)$ is the smallest subset of $V \setminus \{X_i\}$ such that:

$$X_i \perp (V \setminus \{X_i\} \setminus MB(X_i)) \mid MB(X_i)$$

In other words, once we condition on the Markov blanket, X_i becomes independent of all remaining variables in the graph.

Remark 2.2.7 (Markov Blanket in Bayesian Networks). For Bayesian networks, the Markov blanket of a variable X_i consists of three types of variables [3]:

- **Parents:** $pa(X_i) = \{X_j : X_j \rightarrow X_i\}$
- **Children:** $ch(X_i) = \{X_j : X_i \rightarrow X_j\}$
- **Co-parents:** $\{X_k : X_k \rightarrow X_j \leftarrow X_i \text{ for some } X_j \in ch(X_i)\}$

Therefore: $MB(X_i) = pa(X_i) \cup ch(X_i) \cup \{X_k : X_k \rightarrow X_j \leftarrow X_i, X_j \in ch(X_i)\}$.

An important structural relationship emerges when converting the Bayesian network to its moral graph (Def. 2.2.17): the Markov blanket of any variable X_i corresponds exactly to the neighbors of X_i in the resulting undirected moral graph, since the moralization procedure retains all the directed edges while adding undirected edges between co-parents.

2.2.3 Bayesian Hypergraphs

While Bayesian networks and Markov random fields provide powerful frameworks for representing asymmetric and symmetric relationships respectively, many real-world scenarios require the simultaneous representation of both types of dependencies within a single coherent model. Bayesian hypergraphs address this need by combining these two modalities in a unified framework that incorporates both undirected and directed relationships between variables.

The key innovation of Bayesian hypergraphs lies in their use of hyperedges, where both the tail (cause) and head (effect) components can contain arbitrary numbers of variables, rather than being restricted to single variables as in traditional graphs. This generalization enables the representation of complex many-to-many causal relationships that cannot be adequately captured by pairwise edges.

The integration of undirected and directed relationships is achieved through a two-level hierarchical structure. First, variables are coarse-grained into equivalence classes known as chain components, which are determined by the cohead relationships specified in the underlying hyperedges. These chain components are then treated as composite variables that factorize according to a regular Bayesian DAG structure derived from the canonical representation of the hypergraph. This provides the directed, asymmetric relationships at the component level.

Within each chain component, the conditional probability terms from the Bayesian network factorization are further decomposed according to Markov random field structures specified by the hyperedges involved in each conditional probability term. This nested factorization captures the undirected, symmetric relationships within components while preserving the directed relationships between components.

This hybrid structure is particularly well-suited for modeling situations where the collider effect activates only partially. In traditional Bayesian networks, conditioning on a child variable invariably results in conditional dependence among all parent variables. However, we can envision scenarios where conditioning on the child activates dependence only between specific subsets of parents, leaving other parent pairs conditionally independent. Bayesian hypergraphs can encode these nuanced dependency patterns through their Markov random field factorization of conditional probabilities, allowing for more fine-grained control over when and how collider effects manifest.

Definition 2.2.19 (Undirected Hypergraph). An *undirected hypergraph* [4] is a pair $\mathcal{H} = (V, \mathcal{E})$ where:

- V is a finite set of vertices (or nodes)
- $\mathcal{E} = \{E_1, E_2, \dots, E_m\}$ is a collection of *hyperedges*, where each hyperedge $E_i \subseteq V$ is a non-empty subset of vertices

Unlike ordinary graphs where each edge connects exactly two vertices, a hyperedge in a hypergraph can connect any number of vertices simultaneously. When $|E_i| = 2$ for all hyperedges, the hypergraph reduces to an ordinary undirected graph.

A hypergraph is called *k-uniform* if all hyperedges have the same cardinality k , i.e., $|E_i| = k$ for all i . The hypergraph is *simple* if no hyperedge is contained in another hyperedge.

Definition 2.2.20 (Undirected Hypergraph Factorization). A probability distribution over an undirected hypergraph $\mathcal{H} = (V, \mathcal{E})$ has an undirected hypergraph factorization [4] if it can be written as:

$$p(x) = \frac{1}{Z} \prod_{E_i \in \mathcal{E}} \psi_{E_i}(x_{E_i})$$

where:

- The product is over all hyperedges E_i in the hypergraph
- $\psi_{E_i}(x_{E_i}) > 0$ are potential functions defined on the variables x_{E_i} in hyperedge E_i
- Z is the partition function (normalization constant):

$$Z = \sum_x \prod_{E_i \in \mathcal{E}} \psi_{E_i}(x_{E_i})$$

Remark 2.2.8 (Equivalence to Clique-Based Markov Random Field). The undirected hypergraph factorization is equivalent to a Markov random field where each undirected hyperedge E_i is converted into a clique by adding edges between every pair of variables within E_i . Specifically, for each hyperedge $E_i = \{v_1, v_2, \dots, v_k\}$, we add undirected edges $\{v_j, v_\ell\}$ for all $j \neq \ell$, ensuring that E_i forms a complete subgraph (clique) in the resulting graph. The resulting Markov random field, with potential functions defined over these cliques, induces exactly the same conditional independence properties as the original undirected hypergraph factorization.

Definition 2.2.21 (Directed Hypergraph). A *directed hypergraph* [4] is a triple $\mathcal{D} = (V, \mathcal{E}, \phi)$ where:

- V is a finite set of vertices (or nodes)

- $\mathcal{E} = \{e_1, e_2, \dots, e_m\}$ is a collection of *directed hyperedges*
- $\phi : \mathcal{E} \rightarrow 2^V \times 2^V$ is a function that assigns to each hyperedge $e \in \mathcal{E}$ a pair $(T(e), H(e))$ where:
 - $T(e) \subseteq V$ is the *tail* of hyperedge e
 - $H(e) \subseteq V$ is the *head* of hyperedge e
 - $H(e) \cup T(e) = e$ (the hyperedge consists of its head and tail)
 - $H(e) \cap T(e) = \emptyset$ (head and tail are disjoint)

The directed hyperedge e represents a directed relationship from the tail vertices $T(e)$ to the head vertices $H(e)$. When $|T(e)| = |H(e)| = 1$ for all hyperedges, the directed hypergraph reduces to an ordinary directed graph.

A directed hyperedge e with $T(e) = \{v_1, v_2, \dots, v_k\}$ and $H(e) = \{u_1, u_2, \dots, u_\ell\}$ can be denoted as $\{v_1, v_2, \dots, v_k\} \rightarrow \{u_1, u_2, \dots, u_\ell\}$.

Definition 2.2.22 (Cohead and Cotail Relationships). Let $\mathcal{D} = (V, \mathcal{E}, \phi)$ be a directed hypergraph [4]. For any two vertices $u, v \in V$:

- Vertices u and v are *coheads*, denoted $u \sim_H v$, if there exists a directed hyperedge $e \in \mathcal{E}$ such that both $u, v \in H(e)$
- Vertices u and v are *cotails*, denoted $u \sim_T v$, if there exists a directed hyperedge $e \in \mathcal{E}$ such that both $u, v \in T(e)$

In other words:

$$u \sim_H v \Leftrightarrow \exists e \in \mathcal{E} : u, v \in H(e) \quad (2.67)$$

$$u \sim_T v \Leftrightarrow \exists e \in \mathcal{E} : u, v \in T(e) \quad (2.68)$$

Both cohead and cotail are symmetric binary relations on V . Note that these relationships are not necessarily transitive or reflexive.

Remark 2.2.9 (Interpretation of Cohead and Cotail). The cohead relationship captures vertices that serve as joint “targets” or “outputs” of the same directed hyperedge, while the cotail relationship captures vertices that serve as joint “sources” or “inputs” of the same directed hyperedge.

Definition 2.2.23 (Chain Components). Let $\mathcal{D} = (V, \mathcal{E}, \phi)$ be a directed hypergraph [4]. The *chain components* of the hypergraph are the equivalence classes of vertices under the transitive closure of the cohead relationship.

Formally, let $\sim_H^*$ denote the transitive closure of the cohead relation $\sim_H$. That is, $u \sim_H^* v$ if and only if there exists a sequence of vertices $u = w_0, w_1, w_2, \dots, w_k = v$ such that $w_i \sim_H w_{i+1}$ for all $i = 0, 1, \dots, k-1$.

The relation $\sim_H^*$ is an equivalence relation (reflexive, symmetric, and transitive), and thus partitions the vertex set V into disjoint equivalence classes. Each equivalence class $C \subseteq V$ under $\sim_H^*$ is called a *chain component*.

The set of all chain components forms a partition of V :

$$V = \bigcup_i C_i \quad \text{where } C_i \cap C_j = \emptyset \text{ for } i \neq j \quad (2.69)$$

and each C_i is a chain component.

Remark 2.2.10 (Interpretation of Chain Components). Chain components group together vertices that are connected through a sequence of cohead relationships. Within a chain component, any two vertices can be connected via a path where each step involves two vertices appearing together as heads of some directed hyperedge. This captures a form of indirect connectivity in the hypergraph structure that extends beyond immediate cohead relationships.

Definition 2.2.24 (Canonical DAG of a Directed Hypergraph). Let $\mathcal{D} = (V, \mathcal{E}, \phi)$ be a directed hypergraph with chain components $\{C_1, C_2, \dots, C_k\}$ [4]. The *canonical DAG* of $\mathcal{D}$ is a directed acyclic graph $G_{\text{can}} = (V_{\text{can}}, E_{\text{can}})$ where:

- **Vertices:** $V_{\text{can}} = \{C_1, C_2, \dots, C_k\}$ (the chain components serve as vertices)
- **Directed edges:** There is a directed edge $C_i \rightarrow C_j$ in E_{can} if and only if there exist vertices $u \in C_i$ and $v \in C_j$, and a directed hyperedge $e \in \mathcal{E}$ such that:

$$u \in T(e) \text{ and } v \in H(e) \quad (2.70)$$

Formally:

$$E_{\text{can}} = \{(C_i, C_j) : \exists u \in C_i, v \in C_j, e \in \mathcal{E} \text{ such that } u \in T(e) \text{ and } v \in H(e)\} \quad (2.71)$$

The canonical DAG thus captures the high-level causal structure of the directed hypergraph by abstracting individual vertices into their chain components and preserving the directed relationships between components.

Remark 2.2.11 (Properties of the Canonical DAG). The canonical DAG construction ensures that the resulting graph is indeed acyclic. This follows from the fact that vertices within the same chain component are connected only through cohead relationships (not through tail-to-head connections), preventing cycles when the hypergraph structure is projected onto the component level.

Definition 2.2.25 (Bayesian Hypergraph Factorization). Let $\mathcal{H} = (V, \mathcal{E}, \phi)$ be a directed acyclic hypergraph with chain components $\{\tau : \tau \in \mathcal{D}\}$ [4]. A probability distribution P with density f factorizes according to $\mathcal{H}$ if:

1. **Chain component factorization:** The density factorizes across chain components following the structure of the canonical DAG:

$$f(\mathbf{x}) = \prod_{\tau \in \mathcal{D}} f(\mathbf{x}_\tau \mid \mathbf{x}_{\text{pa}(\tau)}) \quad (2.72)$$

where $\mathbf{x}_\tau$ denotes the variables in chain component τ , and $\text{pa}(\tau)$ represents the parent chain components of τ in the canonical DAG.

2. **Within-component factorization:** For each chain component $\tau \in \mathcal{D}$, let $\mathcal{H}_\tau^*$ be the sub-hypergraph containing all hyperedges h such that $H(h) \subseteq \tau$. Then:

$$f(\mathbf{x}_\tau \mid \mathbf{x}_{\text{pa}(\tau)}) = Z^{-1}(\mathbf{x}_{\text{pa}(\tau)}) \prod_{h \in \mathcal{M}(\mathcal{H}_\tau^*)} \psi_h(\mathbf{x}_h) \quad (2.73)$$

where:

- $\mathcal{M}(\mathcal{H}_\tau^*)$ denotes the maximal hyperedges in $\mathcal{H}_\tau^*$
- $\psi_h(\mathbf{x}_h)$ are non-negative potential functions depending only on variables in hyperedge h
- $Z(\mathbf{x}_{\text{pa}(\tau)})$ is the normalization constant ensuring $\int f(\mathbf{x}_\tau \mid \mathbf{x}_{\text{pa}(\tau)}) d\mathbf{x}_\tau = 1$

Remark 2.2.12 (Hybrid Factorization Structure). The Bayesian hypergraph factorization combines two complementary graphical model paradigms [4]:

- **Between chain components:** The factorization follows a Bayesian network structure via the canonical DAG, where each chain component depends only on its parent components, enabling efficient probabilistic inference across the component hierarchy.
- **Within chain components:** The factorization follows a Markov random field structure, where the conditional distribution of variables within a component is expressed as a product of potential functions over maximal hyperedges, capturing complex within-component dependencies.

This hybrid structure allows Bayesian hypergraphs to model both hierarchical causal relationships (through the DAG structure) and complex symmetric interactions (through the MRF structure) within a unified probabilistic framework.

Lemma 2.2.1 (Conditional Independence Preservation in Hypergraph Conversion). *Let $D = (V, E, \phi)$ be a directed acyclic hypergraph with chain components $\{C_1, C_2, \dots, C_k\}$. Define the corresponding undirected hypergraph $H = (V, E^u)$ where each undirected hyperedge $e^u \in E^u$ is the union of the tail and head of some directed hyperedge $e \in E$:*

$$e^u = T(e) \cup H(e) \quad \text{for each } e \in E$$

If a probability distribution P factorizes according to the directed hypergraph D as in 2.2.25, then any conditional independence relationship that holds for the undirected hypergraph factorization also holds for P . Formally, if variables A and B are conditionally independent given C under the undirected hypergraph factorization, then $A \perp B \mid C$ in P .

Proof. Let P factorize according to the directed hypergraph D with the Bayesian hypergraph factorization:

$$f(x) = \prod_{\tau \in \mathcal{D}} f(x_\tau \mid x_{pa(\tau)})$$

where each chain component factorization is:

$$f(x_\tau \mid x_{pa(\tau)}) = Z^{-1}(x_{pa(\tau)}) \prod_{h \in M(H_\tau^*)} \psi_h(x_h)$$

with H_τ^* being the sub-hypergraph containing all hyperedges h such that $H(h) \subseteq \tau$. For the undirected hypergraph H , the factorization would be:

$$f^u(x) = Z_u^{-1} \prod_{e^u \in E^u} \psi_{e^u}(x_{e^u})$$

and we need to show that every potential function $\psi_{e^u}(x_{e^u})$ in the undirected factorization appears in the directed factorization.

Consider any undirected hyperedge $e^u = T(e) \cup H(e)$ for some directed hyperedge $e \in E$. By the definition of chain components, the head $H(e)$ belongs to some chain component $\tau \in \mathcal{D}$. Since e is a directed hyperedge with $H(e) \subseteq \tau$, we have $e \in H_\tau^*$.

Therefore, in the within-component factorization of chain component τ , there exists a potential function $\psi_e(x_e)$ that depends on all variables in the directed hyperedge $e = T(e) \cup H(e) = e^u$.

Since every undirected hyperedge e^u corresponds to some directed hyperedge e , and every directed hyperedge contributes a potential function to the directed hypergraph factorization, the directed factorization contains all the potential functions of the undirected factorization.

By Definition 2.2.25, since the directed hypergraph factorization contains all potential functions from the undirected hypergraph factorization, any conditional independence relationship implied by the undirected hypergraph Markov properties must also hold in the probability distribution P that factorizes according to the directed hypergraph. $\square$

Theorem 2.2.5 (Markov Blanket in Hypergraphs). *Given a directed acyclic hypergraph $D = (V, E, \phi)$ and a variable $X_i \in V$, the Markov blanket of X_i consists of four types of variables [4]:*

- **Parents:** Variables in chain components that are parents of X_i 's chain component in the canonical DAG
- **Children:** Variables in chain components that are children of X_i 's chain component in the canonical DAG

- **Co-tails:** Variables X_j such that $X_i \sim_T X_j$ (i.e., $X_i, X_j \in T(h)$ for some hyperedge h)
- **Co-heads:** Variables X_j such that $X_i \sim_H X_j$ (i.e., $X_i, X_j \in H(h)$ for some hyperedge h)

Remark 2.2.13 (Hypergraph Markov Blanket and Undirected Neighbors). The Markov blanket of a variable X_i in a directed acyclic hypergraph corresponds exactly to the direct neighbors of X_i in the corresponding undirected hypergraph obtained by taking the union of tails and heads for each directed hyperedge. This relationship parallels the connection between Markov blankets in Bayesian networks and neighborhoods in moral graphs, as established in Lemma 2.2.1.

2.3 Partial information decomposition

Having established the theoretical foundations of graphical models for representing causal relationships, we now turn our focus to the partial information decomposition (PID) framework, which provides a complementary perspective on understanding multivariate dependencies through information-theoretic decomposition.

The fundamental setting of PID involves a collection of source variables and a designated target variable, with the goal of decomposing the mutual information between them into interpretable components: unique information (information about the target that can only be derived from a single source variable), synergistic information (information that emerges only when multiple source variables are considered together), and redundant information (information about the target that is shared across multiple source variables and can be obtained from any of them individually).

This decomposition framework allows us to understand the compositional nature of information contributions from each source to the target, enabling us to answer precise questions such as "How much information about target T is derivable from source X_1 but not from any other variable?" or "How much information can be derived from either X_1 or X_2 ?". Unlike unconditional interaction measures that treat all variables symmetrically, the distinction between target and source variables in PID allows us to recover more fine-grained information about the directional nature of interactions between variables, as we get to choose different variables as the target.

Crucially, complete knowledge of all PID components provides sufficient information to derive all conditional mutual information terms in the system. Since conditional independence is mathematically equivalent to vanishing conditional mutual information, this means that PID components contain all the information necessary to determine the conditional independence structure that forms the foundation of graphical model-based causal discovery. This theoretical link between information decomposition and conditional independence structure provides our underlying motivation for investigating the connection between the higher-order information framework of PID and graphical model-based approaches to causality.

Definition 2.3.1 (Partial Information Decomposition Setting). The *Partial Information Decomposition* (PID) framework [5] considers:

- A collection of *source variables* $\mathcal{S} = \{S_1, S_2, \dots, S_n\}$
- A *target variable* T
- A collection of subsets $\mathcal{A} = \{A_1, A_2, \dots, A_m\}$ where each $A_k \subseteq \mathcal{S}$ is a subset of source variables

The framework addresses the fundamental question: How much information about the target T is shared across the subsets $A_1, A_2, \dots, A_m$?

Definition 2.3.2 (Redundancy Measure). A *redundancy measure* [5] is a function $I_\cap$ that quantifies the amount of information about the target variable T that is shared across multiple subsets of source variables.

For a collection of subsets $A_1, A_2, \dots, A_m$ where each $A_k \subseteq \mathcal{S}$, the redundancy measure $I_\cap(A_1, A_2, \dots, A_m; T)$ represents the information about T that is contained in *all* subsets $A_1, A_2, \dots, A_m$ simultaneously.

The redundancy measure aims to capture the intuitive notion of “shared” or “overlapping” information that would be available from any single subset alone, rather than requiring the combination of multiple subsets.

Definition 2.3.3 (Williams-Beer Axioms for Redundancy Measures). Williams and Beer [5] proposed that any valid redundancy measure $I_\cap$ should satisfy the following axioms:

1. **Symmetry:** The redundancy measure is invariant under permutations of the subsets:

$$I_\cap(A_1, A_2, \dots, A_m; T) = I_\cap(A_{\sigma(1)}, A_{\sigma(2)}, \dots, A_{\sigma(m)}; T) \quad (2.74)$$

for any permutation σ .

2. **Self-redundancy:** The redundancy of a subset with itself equals its mutual information with the target:

$$I_\cap(A, A; T) = I(A; T) \quad (2.75)$$

3. **Monotonicity:** Adding more subsets cannot increase the redundancy:

$$I_\cap(A_1, A_2, \dots, A_m, A_{m+1}; T) \leq I_\cap(A_1, A_2, \dots, A_m; T) \quad (2.76)$$

4. **Non-negativity:** Redundancy is always non-negative:

$$I_\cap(A_1, A_2, \dots, A_m; T) \geq 0 \quad (2.77)$$

These axioms formalize the intuitive properties that redundant information should possess while providing constraints for constructing valid redundancy measures.

Remark 2.3.1 (Conceptual Foundation). The Williams-Beer axioms establish the mathematical foundation for measuring shared information [5]. The symmetry axiom ensures that the order of subsets doesn’t matter for redundancy. Self-redundancy establishes that a subset’s redundancy with itself is simply its total information about the target. Monotonicity captures the intuition that adding more constraints (more subsets) can only reduce the amount of shared information. Non-negativity ensures that redundancy represents a meaningful quantity of information.

Definition 2.3.4 (Redundancy Lattice). Given a collection of source subsets $\mathcal{A} = \{A_1, A_2, \dots, A_m\}$ where each $A_k \subseteq \mathcal{S}$, the *redundancy lattice* [5] is a partially ordered set $(\mathcal{L}, \preceq)$ where: **Elements:** The elements of $\mathcal{L}$ are collections of source subsets $\alpha = \{B_1, B_2, \dots, B_k\}$ where each $B_i \subseteq \mathcal{S}$, with the constraint that no subset in α is a superset of another subset in the same collection. Formally, for any $\alpha \in \mathcal{L}$ and any $B_i, B_j \in \alpha$ with $i \neq j$:

$$B_i \not\subseteq B_j \text{ and } B_j \not\subseteq B_i \quad (2.78)$$

This constraint exists because if $B_j \subseteq B_i$, then any information available in B_j is also available in B_i , making B_j redundant in the collection.

Ordering Relation: For elements $\alpha, \beta \in \mathcal{L}$, we define $\alpha \preceq \beta$ if and only if:

$$\forall \mathbf{B} \in \beta, \exists \mathbf{A} \in \alpha \text{ such that } \mathbf{A} \subseteq \mathbf{B} \quad (2.79)$$

The redundancy of element $\alpha = \{A_1, A_2, \dots, A_k\}$ is measured as $I_\cap(A_1, A_2, \dots, A_k; T)$.

Remark 2.3.2 (Intuition for the Ordering Relation). The ordering relation $\alpha \preceq \beta$ captures a fundamental principle about information containment [5].

When $\alpha \preceq \beta$, any redundant information present in β will also be provided in α . This is because:

- If $\beta = \{B_1, B_2, \dots, B_j\}$ and $\alpha = \{A_1, A_2, \dots, A_k\}$ with $\alpha \preceq \beta$, then each B_i contains some A_j as a subset
- Any information that is shared across all sets in β must be available in each B_i
- Since each B_i contains some $A_j \subseteq B_i$, this shared information must also be available in each corresponding A_j
- Therefore, the shared information is also available across all sets in α

This gives us $I_\cap(\alpha; T) \geq I_\cap(\beta; T)$ when $\alpha \preceq \beta$, reflecting that “smaller” collections (in the lattice sense) can only contain more shared information than “larger” collections. The ordering thus captures the containment relationship of redundant information.

Example 2.3.1 (Two-Source Redundancy Lattice). For two source variables S_1, S_2 and target T , consider the subsets $\mathcal{A} = \{\{S_1\}, \{S_2\}, \{S_1, S_2\}\}$. The redundancy lattice contains elements:

- $\alpha_1 = \{\{S_1\}, \{S_2\}\}$ (measuring $I_\cap(\{S_1\}, \{S_2\}; T)$)
- $\alpha_2 = \{\{S_1\}\}$ (measuring $I_\cap(\{S_1\}; T) = I(S_1; T)$)
- $\alpha_3 = \{\{S_2\}\}$ (measuring $I_\cap(\{S_2\}; T) = I(S_2; T)$)
- $\alpha_4 = \{\{S_1, S_2\}\}$ (measuring $I_\cap(\{S_1, S_2\}; T) = I(S_1, S_2; T)$)

The ordering relationships are:

$$\alpha_1 \preceq \alpha_2 \quad (\text{since for } \{S_1\} \in \alpha_2, \text{ we have } \{S_1\} \in \alpha_1 \text{ with } \{S_1\} \subseteq \{S_1\}) \quad (2.80)$$

$$\alpha_1 \preceq \alpha_3 \quad (\text{since for } \{S_2\} \in \alpha_3, \text{ we have } \{S_2\} \in \alpha_1 \text{ with } \{S_2\} \subseteq \{S_2\}) \quad (2.81)$$

$$\alpha_1 \preceq \alpha_4 \quad (\text{since for } \{S_1, S_2\} \in \alpha_4, \text{ both } \{S_1\} \subseteq \{S_1, S_2\} \text{ and } \{S_2\} \subseteq \{S_1, S_2\}) \quad (2.82)$$

$$\alpha_2 \preceq \alpha_4 \quad (\text{since for } \{S_1, S_2\} \in \alpha_4, \text{ we have } \{S_1\} \subseteq \{S_1, S_2\}) \quad (2.83)$$

$$\alpha_3 \preceq \alpha_4 \quad (\text{since for } \{S_1, S_2\} \in \alpha_4, \text{ we have } \{S_2\} \subseteq \{S_1, S_2\}) \quad (2.84)$$

This reflects the information containment hierarchy: the redundant information between S_1 and S_2 (measured by α_1) is contained within the information of each individual source (measured by α_2 and α_3), and all are contained within the information of the combined source (measured by α_4).

Definition 2.3.5 (Partial Information Functions). Given a redundancy lattice $(\mathcal{L}, \preceq)$ and a redundancy measure $I_\cap$, the *partial information functions* $\Pi_{\mathbf{R}}$ are defined implicitly through the relationship [5]:

$$I_\cap(\alpha; T) = \sum_{\beta \preceq \alpha} \Pi_{\mathbf{R}}(\beta; T) \quad (2.85)$$

The partial information functions $\Pi_{\mathbf{R}}(\alpha; T)$ measure the “pure” information contribution of lattice element α that cannot be attributed to any smaller elements in the lattice ordering.

Definition 2.3.6 (Möbius Inversion for Partial Information). The partial information functions can be computed using Möbius inversion on the redundancy lattice [5]. The recursive formula is:

$$\Pi_{\mathbf{R}}(\alpha; T) = I_\cap(\alpha; T) - \sum_{\beta \prec \alpha} \Pi_{\mathbf{R}}(\beta; T) \quad (2.86)$$

where the sum is taken over all elements β that are strictly smaller than α in the lattice ordering ($\beta \prec \alpha$ means $\beta \preceq \alpha$ and $\beta \neq \alpha$). This recursive computation proceeds from the bottom of the lattice upward, ensuring that when computing $\Pi_{\mathbf{R}}(\alpha; T)$, all the values $\Pi_{\mathbf{R}}(\beta; T)$ for $\beta \prec \alpha$ have already been determined.

Definition 2.3.7 (Non-negativity Constraint on Redundancy Measures). For our purposes, we impose an additional constraint on any valid redundancy measure $I_\cap$ beyond the Williams-Beer axioms [5]: the corresponding partial information functions must be non-negative for all lattice elements. Formally, we require:

$$\Pi_{\mathbf{R}}(\alpha; T) \geq 0 \quad \forall \alpha \in \mathcal{L} \quad (2.87)$$

This constraint ensures that each lattice element contributes a meaningful (non-negative) amount of information to the decomposition, preventing pathological cases where some elements would contribute “negative information.”

Remark 2.3.3 (Impossibility Results and Bivariate Restriction). The non-negativity constraint, while intuitively desirable, faces fundamental limitations in the multivariate setting. For two source variables, it is possible to decompose the mutual information into non-negative unique, synergistic, and redundant components. The “Quantifying Unique Information” paper [6] proposed such a measure, often known as BROJA (Bertschinger-Rauh-Olbrich-Jost-Ay).

However, recent impossibility results [7] demonstrate that for three or more source variables, there are no redundancy measures that can always guarantee non-negativity of all PID components. This represents a fundamental mathematical limitation rather than a technical hurdle that can be overcome with better design.

Despite the theoretical challenges posed by multivariate PID non-negativity impossibility results, we circumvent this difficulty by primarily working with bivariate partial information decomposition while still applying it to causal networks involving arbitrary numbers of variables. This approach allows us to leverage the guaranteed non-negativity properties of bivariate PID while maintaining the expressivity needed for multivariate causal analysis. We achieve this by combining collections of variables into single supervariables. For instance, if we have a set of variables V with X_k as the target and we wish to quantify the information contribution of X_i , we group together all variables in $V \setminus \{X_i, X_k\}$ into a single supervariable. The unique information of X_i then represents the information that can only be provided by X_i but not by the supervariable (which contains information from all other variables). Crucially, the patterns of PID most relevant for our causal discovery purposes turn out to be robust to this supervariable transformation, allowing us to bypass the restrictions imposed by the impossibility result.

Definition 2.3.8 (Partial Information Decomposition for Two Sources). For two source variables S_1, S_2 and a target variable T , the *partial information decomposition* [5] decomposes the total mutual information as:

$$I(S_1, S_2; T) = \sum_{\alpha \in \mathcal{L}} \Pi_{\mathbf{R}}(\alpha; T) \quad (2.88)$$

where $\mathcal{L}$ is the redundancy lattice containing the elements: $\alpha_1 = \{\{S_1\}, \{S_2\}\}$, $\alpha_2 = \{\{S_1\}\}$, $\alpha_3 = \{\{S_2\}\}$ and $\alpha_4 = \{\{S_1, S_2\}\}$. Explicitly, this gives:

$$I(S_1, S_2; T) = \Pi_{\mathbf{R}}(\alpha_1; T) + \Pi_{\mathbf{R}}(\alpha_2; T) + \Pi_{\mathbf{R}}(\alpha_3; T) + \Pi_{\mathbf{R}}(\alpha_4; T) \quad (2.89)$$

Definition 2.3.9 (PID Components and Their Interpretations). Each partial information term in the two-source decomposition has a specific interpretation [5]:

1. $\Pi_{\mathbf{R}}(\{\{S_1\}, \{S_2\}\}; T)$ - **Redundant Information**: Information about T that is available from both S_1 and S_2 independently. This represents overlapping information that either source alone can provide.
2. $\Pi_{\mathbf{R}}(\{\{S_1\}\}; T)$ - **Unique Information from S_1** : Information about T that is available only from S_1 and not from S_2 . This is the distinctive contribution of S_1 .
3. $\Pi_{\mathbf{R}}(\{\{S_2\}\}; T)$ - **Unique Information from S_2** : Information about T that is available only from S_2 and not from S_1 . This is the distinctive contribution of S_2 .

4. $\Pi_{\mathbf{R}}(\{\{S_1, S_2\}\}; T)$ - **Synergistic Information**: Information about T that is available only when S_1 and S_2 are considered together, and is not available from either source individually. This represents emergent information from the combination.

Theorem 2.3.1 (Individual Mutual Information Decomposition). *The mutual information between S_1 and the target T can be expressed in terms of PID components [5]:*

$$I(S_1; T) = \Pi_{\mathbf{R}}(\{\{S_1\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_1\}, \{S_2\}\}; T) \quad (2.90)$$

Proof. By the definition of partial information functions, we have:

$$I_{\cap}(\{\{S_1\}\}; T) = \sum_{\beta \preceq \{\{S_1\}\}} \Pi_{\mathbf{R}}(\beta; T) \quad (2.91)$$

Since $I_{\cap}(\{\{S_1\}\}; T) = I(S_1; T)$ by the self-redundancy axiom, and the elements β such that $\beta \preceq \{\{S_1\}\}$ are $\{\{S_1\}\}$ and $\{\{S_1\}, \{S_2\}\}$, we thus have

$$I(S_1; T) = \Pi_{\mathbf{R}}(\{\{S_1\}, \{S_2\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_1\}\}; T). \quad (2.92)$$

This shows that the mutual information between S_1 and T consists exactly of the unique information from S_1 plus the redundant information shared between both sources. $\square$

Theorem 2.3.2 (Conditional Mutual Information Decomposition). *The conditional mutual information between S_2 and the target given S_1 can be expressed in terms of PID components [5]:*

$$I(S_2; T \mid S_1) = \Pi_{\mathbf{R}}(\{\{S_2\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_1, S_2\}\}; T) \quad (2.93)$$

Proof. Using the chain rule of mutual information:

$$I(S_2; T \mid S_1) = I(S_1, S_2; T) - I(S_1; T) \quad (2.94)$$

From the PID decomposition:

$$I(S_1, S_2; T) = \Pi_{\mathbf{R}}(\{\{S_1\}, \{S_2\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_1\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_2\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_1, S_2\}\}; T) \quad (2.95)$$

From the previous theorem:

$$I(S_1; T) = \Pi_{\mathbf{R}}(\{\{S_1\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_1\}, \{S_2\}\}; T) \quad (2.96)$$

Substituting into the chain rule equation:

$$I(S_2; T \mid S_1) = I(S_1, S_2; T) - I(S_1; T) \quad (2.97)$$

$$= [\Pi_{\mathbf{R}}(\{\{S_1\}, \{S_2\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_1\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_2\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_1, S_2\}\}; T)] \quad (2.98)$$

$$- [\Pi_{\mathbf{R}}(\{\{S_1\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_1\}, \{S_2\}\}; T)] \quad (2.99)$$

Cancelling the common terms:

$$I(S_2; T \mid S_1) = \Pi_{\mathbf{R}}(\{\{S_2\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_1, S_2\}\}; T) \quad (2.100)$$

Hence the conditional mutual information $I(S_2; T \mid S_1)$ consists exactly of the unique information from S_2 plus the synergistic information that emerges only when both sources are combined. $\square$

Theorem 2.3.3 (Conditional Mutual Information and Conditional Independence). *For any random variables X , Y , and Z , we have [8]:*

$$I(X; Y \mid Z) = 0 \Leftrightarrow X \perp Y \mid Z \quad (2.101)$$

Proof. Direction 1 ($\Rightarrow$): Assume $I(X; Y | Z) = 0$. We need to show $X \perp Y | Z$.

By definition, conditional mutual information is:

$$I(X; Y | Z) = \mathbb{E}_{X,Y,Z} \left[\log \frac{P(X, Y | Z)}{P(X | Z)P(Y | Z)} \right] \quad (2.102)$$

Since $I(X; Y | Z) = 0$ and the logarithm function ensures this expectation is non-negative (by Jensen's inequality applied to the convex function $-\log$), we have:

$$\log \frac{P(X, Y | Z)}{P(X | Z)P(Y | Z)} = 0 \quad \text{almost surely} \quad (2.103)$$

This implies:

$$\frac{P(X, Y | Z)}{P(X | Z)P(Y | Z)} = 1 \quad \text{almost surely} \quad (2.104)$$

Therefore:

$$P(X, Y | Z) = P(X | Z)P(Y | Z) \quad \text{almost surely} \quad (2.105)$$

which is exactly the definition of conditional independence $X \perp Y | Z$.

Direction 2 ($\Leftarrow$): Assume $X \perp Y | Z$, which means:

$$P(X, Y | Z) = P(X | Z)P(Y | Z) \quad (2.106)$$

Then:

$$I(X; Y | Z) = \mathbb{E}_{X,Y,Z} \left[\log \frac{P(X, Y | Z)}{P(X | Z)P(Y | Z)} \right] \quad (2.107)$$

$$= \mathbb{E}_{X,Y,Z} \left[\log \frac{P(X | Z)P(Y | Z)}{P(X | Z)P(Y | Z)} \right] \quad (2.108)$$

$$= \mathbb{E}_{X,Y,Z} [\log 1] = 0 \quad (2.109)$$

Therefore, $I(X; Y | Z) = 0 \Leftrightarrow X \perp Y | Z$. $\square$

Corollary 2.3.1 (PID Components and Independence). *From the conditional mutual information decomposition and this theorem:*

$$S_2 \perp T | S_1 \Leftrightarrow I(S_2; T | S_1) = 0 \Leftrightarrow \Pi_{\mathbf{R}}(\{\{S_2\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_1, S_2\}\}; T) = 0 \quad (2.110)$$

Since both PID components are non-negative, this occurs if and only if both the unique information from S_2 and the synergistic information are zero.

Theorem 2.3.4 (PID and Markov Equivalence Class Identification). *Under the faithfulness assumption, knowledge of all PID components allows complete identification of the Markov equivalence class of the underlying causal structure.*

Proof. Outline: The argument proceeds through the following logical chain:

- **Markov equivalence through conditional independence:** Two directed acyclic graphs are Markov equivalent if and only if they encode the same set of conditional independence relationships. Under faithfulness, a probability distribution is completely characterized by its conditional independence structure.
- **Conditional independence and conditional mutual information:** From the previous theorem, we have:

$$X \perp Y | Z \Leftrightarrow I(X; Y | Z) = 0 \quad (2.111)$$

Therefore, the complete set of conditional independence relationships is equivalent to knowing which conditional mutual information terms are zero versus non-zero.

- **PID decomposition of conditional mutual information:** As demonstrated, conditional mutual information can be expressed in terms of PID components. For the two-source case:

$$I(S_2; T \mid S_1) = \Pi_{\mathbf{R}}(\{\{S_2\}\}; T) + \Pi_{\mathbf{R}}(\{\{S_1, S_2\}\}; T) \quad (2.112)$$

More generally, any conditional mutual information $I(X; Y \mid Z)$ can be decomposed using appropriate PID components.

- As a result, knowledge of all PID components $\Pi_{\mathbf{R}}(\alpha; T)$ for all lattice elements α provides sufficient information to determine which conditional independence relationships hold, which in turn uniquely identifies the Markov equivalence class of the causal structure under the faithfulness assumption.

□

Lemma 2.3.1 (Conditional Mutual Information and PID Components). *Let S be the set of all source variables, T be the target variable, $C \subseteq S$ be a conditioning set, and $X \in S \setminus C$. Then:*

$$I(X; T \mid C) = I(X, C; T) - I(C; T) = \sum_{\alpha \in \mathcal{A}} \Pi_{\mathbf{R}}(\alpha; T) \quad (2.113)$$

where $\mathcal{A} = \{\alpha \in \mathcal{L} : \alpha \preceq \{\{X, C\}\} \text{ and } \alpha \not\preceq \{\{C\}\}\}$ is the set of all lattice elements that precede $\{\{X, C\}\}$ but do not precede $\{\{C\}\}$.

Proof. : Using the chain rule of mutual information:

$$I(X; T \mid C) = I(X, C; T) - I(C; T) \quad (2.114)$$

By the definition of partial information functions:

$$I(X, C; T) = I_{\cap}(\{\{X, C\}\}; T) = \sum_{\beta \preceq \{\{X, C\}\}} \Pi_{\mathbf{R}}(\beta; T) \quad (2.115)$$

$$I(C; T) = I_{\cap}(\{\{C\}\}; T) = \sum_{\gamma \preceq \{\{C\}\}} \Pi_{\mathbf{R}}(\gamma; T) \quad (2.116)$$

Therefore:

$$I(X; T \mid C) = \sum_{\beta \preceq \{\{X, C\}\}} \Pi_{\mathbf{R}}(\beta; T) - \sum_{\gamma \preceq \{\{C\}\}} \Pi_{\mathbf{R}}(\gamma; T) \quad (2.117)$$

$$= \sum_{\alpha: \alpha \preceq \{\{X, C\}\} \text{ and } \alpha \not\preceq \{\{C\}\}} \Pi_{\mathbf{R}}(\alpha; T) \quad (2.118)$$

The last equality follows because the terms that appear in both sums (i.e., those α such that $\alpha \preceq \{\{X, C\}\}$ and $\alpha \preceq \{\{C\}\}$) cancel out in the subtraction. □

3

Results

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3.1 Unconditional interaction measures for causal discovery

Having introduced interaction measures that leverage unconditional tests to detect higher-order interactions, and established the framework of graphical models and causality, we now explore how these complementary approaches can be combined for causal discovery. Specifically, we investigate the potential of applying unconditional interaction measures—traditionally used for detecting statistical dependencies—to the problem of inferring causal structure from observational data.

While unconditional tests are inherently less powerful than conditional independence tests, they offer significant computational advantages that scale more favorably with the number of variables. This motivates a systematic investigation: Can we harness the computational efficiency of unconditional tests for causal discovery, and if so, what are the fundamental limitations we must account for? Which causal properties are detectable through unconditional tests alone, and which ones fundamentally require conditional testing? Once we understand these constraints, how can we develop principled procedures that extract maximal causal information from unconditional tests while respecting their theoretical limitations?

To address these questions systematically, we begin by examining the application of the Lancaster interaction measure for V-structure detection—a known technique for which we provide an original proof. We then demonstrate the fundamental limitations of unconditional tests in causal

discovery, starting with comprehensive analysis of the three-variable case and subsequently generalizing to extensions and augmentations of three-variable structures. Through this analysis, we establish which causal configurations can and cannot be distinguished using only unconditional independence information.

Once these patterns are thoroughly understood, we formalize these limitations by introducing a coarse-graining procedure that partitions variables into equivalence classes, where variables within the same equivalence class represent exactly those whose causal roles cannot be distinguished using unconditional tests. This equivalence class structure provides a precise mathematical characterization of the information that unconditional tests can and cannot extract about causal relationships.

Finally, we synthesize all of these theoretical results to produce a comprehensive procedure that utilizes unconditional tests to their fullest extent, maximizing the causal information that can be reliably extracted while respecting the fundamental limitations we have identified. This procedure bridges the gap between the computational efficiency of unconditional tests and the theoretical rigor required for valid causal inference.

3.1.1 Lancaster Interaction measure for V-structure detection

Definition 3.1.1 (V-Structure Detection via Lancaster Interaction Measure). The Lancaster interaction measure provides a method for detecting V-structures (colliders) among three variables [1]. The procedure involves two stages:

1. **Marginal independence test:** Test the null hypothesis $H_0^{(1)} : X \perp Y$
2. **Conditional dependence tests:** If $X \perp Y$ is not rejected, test the three factorization hypotheses:

$$H_0^{(2a)} : X \perp (Y, Z) \quad (\text{equivalently } (X, Y, Z) = P_X P_{YZ}) \quad (3.1)$$

$$H_0^{(2b)} : Y \perp (X, Z) \quad (\text{equivalently } P_{XYZ} = P_Y P_{XZ}) \quad (3.2)$$

$$H_0^{(2c)} : Z \perp (X, Y) \quad (\text{equivalently } P_{XYZ} = P_Z P_{XY}) \quad (3.3)$$

A V-structure $X \rightarrow Z \leftarrow Y$ is detected when the marginal independence $X \perp Y$ holds, but all three conditional independence relationships are rejected.

We now prove a result that was stated in Ref. [1] without proof and for which, to the best of our knowledge, a proof is not found in the literature. Here we provide a detailed original proof.

Theorem 3.1.1 (V-Structure Detection Consistency). *Assume that the probability distribution P is faithful to the underlying causal graph G , for three random variables X , Y , and Z , the following statements are equivalent:*

1. *The causal structure is a V-structure $X \rightarrow Z \leftarrow Y$*
2. *$X \perp Y$ (marginal independence) AND all three factorization hypotheses are rejected: $X \not\perp (Y, Z)$, $Y \not\perp (X, Z)$, and $Z \not\perp (X, Y)$*

Proof. Direction 1: If the structure is $X \rightarrow Z \leftarrow Y$, then the conditions hold.

From the V-structure factorization $P(X, Y, Z) = P(X)P(Y)P(Z | X, Y)$:

- **Marginal independence:**

$$P(X, Y) = \sum_z P(X, Y, Z) \quad (3.4)$$

$$= \sum_z P(X)P(Y)P(Z | X, Y) \quad (3.5)$$

$$= P(X)P(Y) \sum_z P(Z | X, Y) \quad (3.6)$$

$$= P(X)P(Y) \quad (3.7)$$

Therefore: $X \perp Y \checkmark$

- **Rejection of factorizations:**

- $X \not\perp (Y, Z)$: If this were true, then $P(X, Y, Z) = P(X)P(Y, Z)$, but this contradicts the V-structure factorization unless $P(Z | X, Y) = P(Z | Y)$, which would mean X doesn't influence Z .
- Similarly for $Y \not\perp (X, Z)$ and $Z \not\perp (X, Y)$.

Direction 2: If the conditions hold, then the structure must be $X \rightarrow Z \leftarrow Y$.

Given conditions: $X \perp Y$ but $X \not\perp (Y, Z)$, $Y \not\perp (X, Z)$, and $Z \not\perp (X, Y)$.

- **Apply contrapositive of contraction axiom:** Taking the contrapositive of the contraction axiom we have $X \not\perp (Y, Z) \implies X \not\perp Y$ or $X \not\perp Z | Y$. Since we know $X \perp Y$, we cannot have $X \perp Z | Y$. Hence $X \not\perp Z | Y$.
- **Apply similar argument for Y:** Since $Y \not\perp (X, Z)$ and $X \perp Y$, we get: $Y \not\perp Z | X$.
- **Apply d-separation to marginal independence:** By d-separation principles, $X \perp Y$ implies that all paths from X to Y are blocked when no variables are conditioned upon.
- **Identify unblocked path through Z:** Since $X \not\perp Z | Y$ and $Y \not\perp Z | X$, there must be a path from X to Y through Z , where the path from X to Z is unblocked, and the path from Z to Y is unblocked.
- **Rule out chain and confounder paths:** Note that the path cannot be of the form $X \rightarrow \dots \rightarrow Y \rightarrow \dots \rightarrow Z$ or $Y \rightarrow \dots \rightarrow X \rightarrow \dots \rightarrow Z$, since if such a path were open, that would imply an unblocked path from X to Y (ignoring Z), which contradicts our assumption that $X \perp Y$. In particular, we must have a path of the form $X \dots Z \dots Y$ which is blocked by the absence of conditioning on Z .
- **Establish Z as collider:** By d-separation, Z has to be a collider, since any other structure (chain or confounder involving Z) would imply $X \not\perp Y$, contradicting our assumptions.
- **Conclude V-structure:** The collider structure is the only possible structure that is consistent with unconditional independence of X and Y but conditional dependence in all other factorizations. Therefore, the only possible graphical structure is the V-structure $X \rightarrow Z \leftarrow Y$.

□

Corollary 3.1.1. *Under the faithfulness assumption in Definition 2.2.10, testing for marginal independence followed by rejection of all factorization hypotheses provides a necessary and sufficient condition for V-structure detection using only unconditional independence tests.*

3.1.2 Limitation of unconditional tests in three variable structures

Although we have shown that unconditional tests are able to conclude the V-structure above, there are fundamental limitations to the power of unconditional tests for causal discovery.

Theorem 3.1.2 (Limitations of Unconditional Independence Tests for Structural Discovery). *Consider three random variables A , B , and C with a probability distribution that is faithful to the underlying causal graph. For the chain structure $A \rightarrow B \rightarrow C$ and the confounder structure $A \leftarrow B \rightarrow C$, the following limitation holds:*

Unconditional Equivalence: *All variables exhibit identical unconditional independence behavior:*

$$A \not\perp B, \quad A \not\perp C, \quad B \not\perp C, \quad A \not\perp (B, C), \quad B \not\perp (A, C), \quad C \not\perp (A, B) \quad (3.8)$$

In particular, we can arbitrarily permute the variables A, B, C and the unconditional dependence relationship will remain invariant.

Conditional Distinguishability: *The middle variable B distinct conditional independence signatures that distinguishes it from the other variables:*

$$\text{Chain } (A \rightarrow B \rightarrow C) : \quad A \perp C \mid B \text{ but } A \not\perp B \mid C \text{ and } C \not\perp B \mid A \quad (3.9)$$

$$\text{Confounder } (A \leftarrow B \rightarrow C) : \quad A \perp C \mid B \text{ but } A \not\perp B \mid C \text{ and } C \not\perp B \mid A \quad (3.10)$$

This shows that while both structures can be distinguished from the V-structure using unconditional tests (since V-structure has $A \perp C$ marginally), unconditional independence tests alone cannot identify the variable plays the special “mediating” or “confounding” role, which makes it weaker compared to conditional independence queries

In the Theorem below, we summarise the capabilities of unconditional tests in terms of DAG identification for $d = 3$.

Theorem 3.1.3 (Unconditional Independence Tests for Three-Variable Structures). *Consider three random variables A , B , and C with a probability distribution that is faithful to the underlying causal graph. Unconditional independence tests can uniquely determine the following Markov equivalence classes:*

Case 1: Single Variable Independence. *If one variable is independent of the other two variables, the unconditional independence pattern uniquely determines the structure up to Markov equivalence.*

Specifically, if A is independent of both B and C :

$$A \perp B, \quad A \perp C, \quad A \perp (B, C), \quad (3.11)$$

then the only possible structures are those where A is disconnected from both B and C , while B and C may or may not be connected:

- $A \quad B - C$ (A isolated, B and C connected)
- $A \quad B \quad C$ (all three variables mutually independent)

The relationship between B and C can then be determined by testing $B \perp C$:

$$B \perp C \Rightarrow \text{Structure: } A \quad B \quad C \text{ (complete independence)} \quad (3.12)$$

$$B \not\perp C \Rightarrow \text{Structure: } A \quad B - C \text{ (} A \text{ isolated, } B\text{-}C \text{ connected)} \quad (3.13)$$

Case 2: V-Structure Detection. As shown above, the collider structure can be detected by the Lancaster measure through the following unconditional tests:

$$A \perp C, \quad A \not\perp B, \quad B \not\perp C, \quad A \not\perp (B, C), \quad B \not\perp (A, C), \quad C \not\perp (A, B) \quad (3.14)$$

Case 3: Complete Dependence. If all unconditional dependencies hold:

$$A \not\perp B, \quad A \not\perp C, \quad B \not\perp C, \quad A \not\perp (B, C), \quad B \not\perp (A, C), \quad C \not\perp (A, B), \quad (3.15)$$

then the structure belongs to the Markov equivalence class containing both the chain $A \rightarrow B \rightarrow C$ and the confounder $A \leftarrow B \rightarrow C$, which cannot be distinguished using unconditional tests alone.

Proof. **Case 1** can easily be checked by enumerating all possible DAG structures that satisfy the unconditional independence; **Case 2** follows directly from Theorem 3.1.1; **Case 3** follows directly from Theorem 3.1.2. To show completeness, note that **Case 2** and **Case 3** covers all connected three variable structures in definition 2.2.7, while **Case 1** covers all DAG where at least one of the variables is isolated from the rest. $\square$

Conclusion: For three-variable structures, unconditional independence tests can distinguish between three fundamental classes of causal configurations: structures with isolated variables where at least one variable is independent from the other two, V-structures (colliders) where two independent causes converge on a common effect, and complete dependence structures encompassing both chain and confounder configurations where all variables exhibit pairwise dependencies.

3.1.3 Limitations of unconditional tests for extended structures

Now that we have established a complete taxonomy of three-variable structures detectable by unconditional tests, we turn to examining the limitations of unconditional tests when dealing with extensions and generalizations of these basic three-variable patterns.

Theorem 3.1.4 (Unconditional Independence Limitations for Extended Structures). *Assume that the probability distribution P is faithful to the underlying causal graph G . For structures extending beyond the basic three-variable patterns, unconditional independence tests exhibit the following limitations and capabilities:*

Part I (Collider Extension Ambiguity): If variables A , B , and C form a collider structure $A \rightarrow C \leftarrow B$, then any descendant D of C will exhibit the same unconditional independence relationships with A and B as C itself:

$$A \not\perp (B, C), \quad A \not\perp C, \quad B \not\perp C \quad \Leftrightarrow \quad A \not\perp (B, D), \quad A \not\perp D, \quad B \not\perp D \quad (3.16)$$

Therefore, unconditional tests alone cannot distinguish between:

$$\begin{array}{ccc} A \longrightarrow C \longleftarrow B & & A \longrightarrow D \longleftarrow B \\ \downarrow & & \downarrow \\ D & \text{and} & C \end{array}$$

Part II (Chain Extension Ambiguity): If we have a V-structure of the form $A \rightarrow C \leftarrow B$, then ancestors of A exhibit identical unconditional independence properties as A . For the structure $D \rightarrow A \rightarrow C \leftarrow B$, the variables D and A have the same unconditional independence relationships with respect to B and C :

$$A \perp B, \quad A \not\perp C, \quad B \not\perp C \quad (\text{for } A \text{ in } A \rightarrow C \leftarrow B) \quad (3.17)$$

$$D \perp B, \quad D \not\perp C, \quad B \not\perp C \quad (\text{for } D \text{ in } D \rightarrow A \rightarrow C \leftarrow B) \quad (3.18)$$

Therefore, unconditional tests alone cannot distinguish between

$$D \rightarrow A \rightarrow C \leftarrow B \quad \text{and} \quad A \rightarrow D \rightarrow C \leftarrow B.$$

However, we can distinguish D from being a child of C because $D \not\perp B$ would be required if D were a child of C in the structure $A \rightarrow C \leftarrow B$ with $C \rightarrow D$.

Part III (Diamond Structure Detection): Unconditional tests can rule out diamond structures. For a diamond structure $A \leftarrow D \rightarrow B$ combined with $A \rightarrow C \leftarrow B$, the pairwise independence $A \perp B$ would be violated, preventing misidentification as a simple collider:

$$A \not\perp B \quad (\text{due to diamond path } A \leftarrow D \rightarrow B) \quad (3.19)$$

Therefore, if we observe $A \perp B$ along with the collider pattern, we can rule out diamond structures.

Proof. The unconditional independence properties follows straightforwardly through the application of d-separation with an empty conditioning set: For instance, in chain extension ambiguity $D \rightarrow A \rightarrow C \leftarrow B$, the directed path $D \rightarrow A \rightarrow C$ implies that $D \not\perp C$, while $D \perp B$ is justified by the blocked segment $A \rightarrow C \leftarrow B$ in the path given the empty. $\square$

Conclusion: Unconditional independence tests can detect the collider structure and distinguish the role that each variable plays within the collider structure. However, they cannot distinguish between:

1. A collider and its descendants
2. Different positions in a chain when they have equivalent path relationships

3.1.4 Unconditional Independence Equivalence Class

Motivated by the observation that the indistinguishability of certain causal structures by unconditional tests stems from variables exhibiting identical unconditional independence behaviors, we introduce a formal notion of unconditional independence equivalence classes to crisply characterize the fundamental limits of unconditional tests for causal discovery.

Definition 3.1.2 (Unconditional Independence Equivalence Class). Let $V = \{X_1, X_2, \dots, X_n\}$ be a collection of random variables, and let us define an equivalence relation $\sim$ on V as follows:

For variables $X_i, X_j \in V$ with $i \neq j$, we say $X_i \sim X_j$ if and only if the following two conditions hold:

1. **Unconditional dependence:** $X_i \not\perp X_j$
2. **Identical unconditional profiles:** For any subset $K_1, K_2 \subseteq V \setminus \{X_i, X_j\}$:

$$K_1, X_i \perp K_2 \Leftrightarrow K_1, X_j \perp K_2 \quad (3.20)$$

Then the *unconditional independence equivalence class* of a variable X_i is the set:

$$[X_i] = \{X_j \in V : X_j \sim X_i\} \quad (3.21)$$

This relation partitions the variable set V into disjoint equivalence classes where variables within each class are unconditionally indistinguishable in terms of their independence relationships with all other subsets of variables.

Remark 3.1.1 (Verification of Equivalence Relation). We verify that $\sim$ is an equivalence relation:

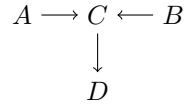
Reflexivity: $X_i \sim X_i$ holds trivially since $X_i \not\perp X_i$ (a variable is always dependent on itself) and $K_1, X_i \perp K_2 \Leftrightarrow K_1, X_j \perp K_2$ for any K .

Symmetry: If $X_i \sim X_j$, then $X_i \not\perp X_j \Leftrightarrow X_j \not\perp X_i$ (by symmetry of dependence) and the identical unconditional profiles condition is symmetric by definition.

Transitivity: If $X_i \sim X_j$ and $X_j \sim X_k$, then:

- $X_i \not\perp X_j$ and $X_j \not\perp X_k$ imply $X_i \not\perp X_k$ (since if $X_i \perp X_k$, the identical profiles would require $X_j \perp X_k$, contradiction)
- For any $K_1, K_2 \subseteq V \setminus \{X_i, X_k\}$: $K_1, X_i \perp K_2 \Leftrightarrow K_1, X_j \perp K_2 \Leftrightarrow K_1, X_k \perp K_2$

Example 3.1.1 (Collider with Descendants). Consider the following structure:



In this diagram, variables C and D belong to the same equivalence class because:

- $C \not\perp D$ (direct causal connection)
- For any subset K not containing C or D :

$$C \perp \{A\} \Leftrightarrow D \perp \{A\} \quad (\text{both false}) \quad (3.22)$$

$$C \perp \{B\} \Leftrightarrow D \perp \{B\} \quad (\text{both false}) \quad (3.23)$$

$$C \perp \{A, B\} \Leftrightarrow D \perp \{A, B\} \quad (\text{both false}) \quad (3.24)$$

$$A \perp \{C, B\} \Leftrightarrow A \perp \{D, B\} \quad (\text{both false}) \quad (3.25)$$

This formalizes why unconditional independence tests cannot distinguish between C and D in this structure.

Definition 3.1.3 (Independence of Equivalence Classes). Let $\tau_1, \tau_2, \dots, \tau_n$ be unconditional independence equivalence classes. We say that:

$$\tau_1, \tau_2, \dots, \tau_i \perp \tau_{i+1}, \dots, \tau_n \quad (3.26)$$

if and only if for any sequence of variables $X_1, X_2, \dots, X_n$ with $X_j \in \tau_j$ for all $j \in \{1, 2, \dots, n\}$, the following independence statement holds:

$$X_1, X_2, \dots, X_i \perp X_{i+1}, \dots, X_n \quad (3.27)$$

In other words, independence between equivalence classes means that independence holds for any choice of representatives from those classes.

Theorem 3.1.5 (Unconditional Independence Preservation under Equivalence Class Mapping). Let $V = \{X_1, X_2, \dots, X_n\}$ be a collection of random variables with unconditional independence equivalence classes. Let $f : V \rightarrow \{\text{equivalence classes}\}$ be the mapping that assigns each variable to its equivalence class, i.e., $f(X_i) = [X_i]$.

For any disjoint subsets $A, B \subseteq V$, the following equivalence holds:

$$A \perp B \Leftrightarrow \{f(X_i) : X_i \in A\} \perp \{f(X_j) : X_j \in B\} \quad (3.28)$$

where the right-hand side uses the definition of independence between equivalence classes.

Proof. Direction 1 ($\Rightarrow$): Assume $A \perp B$. We need to show $\{f(X_i) : X_i \in A\} \perp \{f(X_j) : X_j \in B\}$.

By the definition of equivalence class independence, we need to show that for any choice of representatives $Y_i \in f(X_i)$ for $X_i \in A$ and $Z_j \in f(X_j)$ for $X_j \in B$, we have:

$$\{Y_i : X_i \in A\} \perp \{Z_j : X_j \in B\} \quad (3.29)$$

Let $Y_i \in f(X_i) = [X_i]$ and $Z_j \in f(X_j) = [X_j]$ be arbitrary representatives. Since $Y_i \sim X_i$, by the identical unconditional profiles property with $K_1 = \{Y_k : X_k \in A, k \neq i\}$ and $K_2 = \{Z_j : X_j \in B\}$:

$$K_1, Y_i \perp K_2 \Leftrightarrow K_1, X_i \perp K_2 \quad (3.30)$$

Note that $K_1 \cup K_2 \subseteq V \setminus \{X_i, Y_i\}$ as required by the equivalence relation definition.

Similarly, for each $Z_j \sim X_j$, using the identical unconditional profiles property with $K_1 = \{Y_k : X_k \in A\}$ and $K_2 = \{Z_k : X_k \in B, k \neq j\}$, we have:

$$K_1 \perp K_2, Z_j \Leftrightarrow K_1 \perp K_2, X_j \quad (3.31)$$

By applying this equivalence iteratively for each representative and using the fact that $A \perp B$ (which means every variable in A is independent of every variable in B), we can establish that:

$$\{Y_i : X_i \in A\} \perp \{Z_j : X_j \in B\} \quad (3.32)$$

The key insight is that the identical unconditional profiles property ensures that when we replace any variable with an equivalent one, the independence relationships with all other subsets (including the other variables in our independence statement) are preserved.

Direction 2 ($\Leftarrow$): Assume $\{f(X_i) : X_i \in A\} \perp \{f(X_j) : X_j \in B\}$.

By the definition of equivalence class independence, this means that for any representatives $Y_i \in [X_i]$ and $Z_j \in [X_j]$:

$$\{Y_i : X_i \in A\} \perp \{Z_j : X_j \in B\} \quad (3.33)$$

In particular, choosing $Y_i = X_i$ and $Z_j = X_j$ as representatives, we get:

$$A \perp B \quad (3.34)$$

□

Conclusion: An unconditional independence statement holds between sets of variables if and only if the corresponding statement holds between their equivalence classes under the equivalence class independence definition.

This allows us to identify the boundary between the causal roles that can or cannot be distinguished through unconditional tests. Specifically, variables within the same equivalence class exhibit identical unconditional independence behaviors and hence cannot be further distinguished, whereas variables across different equivalence classes can be differentiated through their distinct unconditional patterns. In turn, this theoretical characterization serves as the foundation for our final objective of developing a systematic procedure for causal discovery that leverages unconditional tests to their full theoretical limits while respecting these fundamental constraints.

3.1.5 Unconditional tests for causal discovery via hypergraphs

Finally, we synthesize all of the previous results to produce an algorithm that utilizes unconditional tests to their fullest extent for causal discovery. In particular, we leverage the co-head structures from Bayesian hypergraphs to model the coarse-grained causal relationships that can

be reliably identified while respecting the fundamental limitations imposed by equivalence class indistinguishability.

Theorem 3.1.6 (Unconditional tests for causal discovery via hypergraphs). *Given a set of variables V and a joint probability distribution that is faithful to an underlying directed acyclic hypergraph, the following procedure determines all causal structure detectable by unconditional independence tests:*

Construction Algorithm:

1. **Equivalence Class Formation:** Partition variables V into unconditional independence equivalence classes $\{\tau_1, \tau_2, \dots, \tau_k\}$ as defined in Definition 3.1.3.
2. **Inter-Class Structure Discovery:** For all triples of equivalence classes $\{\tau_i, \tau_j, \tau_k\}$, apply the three-variable unconditional tests from Definition 3.1.3 to establish the causal structure between chain components.
3. **Hypergraph Construction:** For each equivalence class τ_i :
 - Create a single directed hyperedge h_i with head $H(h_i) = \tau_i$
 - Set the tail $T(h_i)$ to contain all parent equivalence classes of τ_i identified by the unconditional tests

This construction determines all causal structure that can be detected using unconditional independence tests under the faithfulness assumption.

Proof. We prove that this procedure captures all detectable causal structure by showing completeness across and within equivalence classes.

Sufficiency of Equivalence Class Analysis: By Theorem 3.1.5, any unconditional independence relationship between sets of variables is equivalent to the corresponding relationship between their equivalence classes. Therefore, detecting causal structure across equivalence classes is sufficient, as variables within each equivalence class exhibit identical unconditional independence behavior.

Intra-Class Structure: Causal structures within equivalence classes cannot be distinguished by unconditional tests, since all variables in the same class have identical unconditional independence profiles by definition. The cohead construction captures this indistinguishability while preserving the shared causal role of these variables.

Inter-Class Relationship Determination: Under faithfulness, it suffices to determine all pairwise relationships between equivalence classes. For any two equivalence classes A and B , we have two cases:

Case 1: A and B are unconditionally independent. Either:

- There exists a third equivalence class C that is unconditionally dependent with both A and B , implying a V-structure $A \rightarrow C \leftarrow B$ (detected by three-variable tests)
- No such C exists, meaning A and B are causally isolated

Case 2: A and B are unconditionally dependent. There must exist a third equivalence class C such that A and B have different unconditional independence behaviors with respect to C (one is independent while the other is not). This asymmetry allows us to detect V-structure using the three-variable unconditional tests.

Completeness: Since every pairwise relationship between equivalence classes can be determined through three-variable tests, and intra-class structure is captured by the co-head construction, this procedure determines all causal structure detectable by unconditional independence tests. $\square$

Remark 3.1.2 (Factorization Consistency). The hypergraph construction satisfies desirable factorization properties that ensure consistency with the underlying causal structure. Since co-head relationships are specified only within equivalence classes, and equivalence classes are disjoint by construction, the chain components of the resulting hypergraph are exactly the equivalence classes $\{\tau_1, \tau_2, \dots, \tau_k\}$.

By the Bayesian hypergraph factorization (Definition 2.2.25), the chain components factor according to a regular Bayesian DAG structure:

$$f(x) = \prod_{i=1}^k f(x_{\tau_i} \mid x_{pa(\tau_i)}) \quad (3.35)$$

This is precisely the factorization structure that the unconditional tests identify between equivalence classes. The fact that variables within the same equivalence class share identical unconditional independence properties ensures consistency between the causal properties of the equivalence class as a unit and the underlying individual variables, validating the hypergraph representation as a faithful abstraction of the detectable causal structure.

3.2 PID type signatures of causal structures

Our introduction of PID has presented the theoretical motivation for exploring the relationship between information decomposition and causality; namely, that knowledge of all PID components allows us to recover all conditional mutual information terms, which in turn enables us to determine the complete conditional independence structure. While we have explored the limitations of unconditional tests with respect to causal discovery, partial information decomposition differs fundamentally from unconditional interaction measures in that it incorporates the notion of a target variable that is treated asymmetrically compared to all other variables. This asymmetric treatment hints toward greater potential for PID in causal discovery applications.

One of our primary motivations for using PID to understand causality is to achieve a “localist” interpretation of various causal roles. The typical framework of causal discovery involves using conditional independence queries on the underlying probability distribution to identify an equivalence class of consistent graphical models—a fundamentally global approach that operates on the joint distribution and considers the entire space of possible graphical models simultaneously. In contrast, a local approach enables us to focus on one specific part of the causal graph, examining a particular variable and characterizing the structure of its immediate neighborhood.

PID is especially well-suited for this localist perspective, as we can designate any variable of interest as the target and systematically analyze how other variables contribute information to that target. We expect that variables with direct causal relationships to the target will exhibit characteristic PID signatures that allow us to identify their specific causal roles. For instance, causal parents should contribute unique information that cannot be obtained from other sources, while co-parents should exhibit synergistic information patterns that reflect their shared influence through common children.

In this section, we explore the distinctive PID signatures of direct causal neighbors and co-parents, introducing appropriate theoretical conditions to establish a bidirectional correspondence between causal roles and information-theoretic signatures. We demonstrate that under suitable assumptions such as faithfulness and persistent relevance, specific PID patterns both characterize and are characterized by particular causal relationships. Furthermore, we generalize these results to the hypergraph setting, where the richer structural relationships enable more nuanced PID signatures for complex causal configurations.

We begin with a theorem that establishes the fundamental relationship between conditional mutual information and unique information, namely that under appropriate positivity assumptions,

the vanishing of any conditional mutual information implies zero unique information. This theorem serves as a crucial building block for establishing the connection between causal direct neighbors and unique information patterns. The underlying logic is straightforward: for any variable that is not a direct causal neighbor of the target, we can construct some conditioning set that renders their conditional mutual information zero, and through this theorem we can then establish that such variables contribute zero unique information to the target.

Theorem 3.2.1 (Conditional Independence and Unique Information). *Let S be the set of all source variables, $C \subseteq S$ be a conditioning set, and $X_i, X_j \in S$ with $X_i, X_j \notin C$. Given the positivity assumption, if there exists a conditioning set C such that $I(X_i; X_j | C) = 0$, then X_j has zero unique information when X_i is set as the target variable.*

Formally, if $I(X_i; X_j | C) = 0$, then $\Pi_{\mathbf{R}}(\{\{X_j\}\}; X_i) = 0$.

Proof. From Lemma 2.3.1, we have:

$$I(X_j; X_i | C) = \sum_{\alpha \in \mathcal{A}} \Pi_{\mathbf{R}}(\alpha; X_i) \quad (3.36)$$

where $\mathcal{A} = \{\alpha \in \mathcal{L} : \alpha \preceq \{\{X_j, C\}\} \text{ and } \alpha \not\preceq \{\{C\}\}\}$.

By assumption, $I(X_i; X_j | C) = 0$, so:

$$\sum_{\alpha \in \mathcal{A}} \Pi_{\mathbf{R}}(\alpha; X_i) = 0 \quad (3.37)$$

The set $\mathcal{A}$ contains lattice elements that involve X_j but are not entirely determined by C . Crucially, this set includes $\{\{X_j\}\}$ since:

- $\{\{X_j\}\} \preceq \{\{X_j, C\}\}$ (as $\{X_j\} \subseteq \{X_j, C\}$)
- $\{\{X_j\}\} \not\preceq \{\{C\}\}$ (since $X_j \notin C$, we have $\{X_j\} \not\subseteq C$)

Therefore, $\{\{X_j\}\} \in \mathcal{A}$, and the sum includes $\Pi_{\mathbf{R}}(\{\{X_j\}\}; X_i)$.

Since all PID components are non-negative by our constraint on redundancy measures:

$$\Pi_{\mathbf{R}}(\alpha; X_i) \geq 0 \quad \forall \alpha \in \mathcal{L} \quad (3.38)$$

and the sum of non-negative terms equals zero, each individual term must be zero:

$$\Pi_{\mathbf{R}}(\alpha; X_i) = 0 \quad \forall \alpha \in \mathcal{A} \quad (3.39)$$

Therefore, X_j has zero unique information about target X_i . $\square$

Definition 3.2.1 (Source-dependent PID). For a given set of source variables $S = \{S_1, S_2, \dots, S_n\}$ and target variable T , we denote $\Pi_R^S(\alpha; T)$ as the PID component corresponding to lattice element α when S is the source set.

Remark 3.2.1 (Source Set Dependency of PID Components). The explicit notation Π_R^S emphasizes that PID components are fundamentally dependent on the choice of source variables. When we increase the number of source variables from S to $S' \supset S$, we increase the number of lattice elements that precede any particular element in the redundancy lattice ordering.

This has important implications for the interpretation of unique information. The unique information of a variable X in bivariate PID, denoted $\Pi_R^{\{X, Y\}}(\{\{X\}\}; T)$, may include information that X redundantly shares with some variable $Z \notin \{X, Y\}$. Once we expand the source set to include Z , this previously “unique” information is reclassified as redundant information $\Pi_R^{\{X, Y, Z\}}(\{\{X\}, \{Z\}\}; T)$.

Therefore, unique information is only meaningful relative to the variables under consideration, and expanding the source set can reveal hidden redundancies that were not apparent in smaller variable sets.

Assumption 3.2.1 (Unique Information Monotonicity Property). A redundancy measure $I_\cap$ satisfies the unique information monotonicity property if the following holds: Let T be the target variable and X_1, X_2 be source variables. For any additional variable C , let $X'_2 = \{X_2\} \cup \{C\}$ be the supervariable combining C and X_2 . Then we require:

$$\Pi_R^{\{X_1, X'_2\}}(\{\{X_1\}\}; T) \leq \Pi_R^{\{X_1, X_2\}}(\{\{X_1\}\}; T) \quad (3.40)$$

In other words, augmenting one source variable with additional information cannot increase the unique information contribution of the other source variable.

Remark 3.2.2 (Intuition for Unique Information Monotonicity). This property captures the intuitive notion that unique information represents information that cannot be retrieved from other sources. Since the unique information of X_1 measures what X_1 provides about the target T that X_2 cannot provide, adding strictly more information to X_2 (by forming X'_2) should not increase the unique contribution of X_1 .

The augmented variable X'_2 contains all the information that X_2 originally had, plus additional information from C . Therefore, any information that was previously unique to X_1 (relative to X_2) may now be partially or fully captured by X'_2 , but no new unique information for X_1 should emerge from this augmentation. This monotonicity property ensures that the notion of “uniqueness” remains coherent as we expand the information available to competing sources.

Assumption 3.2.2 (Persistent Relevance Property). A joint probability distribution $P(S, T)$ of the sources S and target T satisfy the *Persistent Relevance Property* under the redundancy measure $I_\cap$ if for any variable $X \in S$ and target T :

$$(\forall C \subseteq S \setminus \{X\} : I(X; T \mid C) > 0) \Rightarrow \Pi_R(\{\{X\}\}; T) > 0 \quad (3.41)$$

In other words, if a variable X maintains positive conditional mutual information with the target T regardless of what other variables are conditioned upon, then X must possess positive unique information about T .

Remark 3.2.3 (Intuition and Motivation). The Persistent Relevance Property addresses potential pathological cases in PID where a variable might always maintain some dependency with the target, yet contribute zero unique information.

To see this, consider a scenario where two variables X and Y have:

- Positive redundant information: $\Pi_R(\{\{X\}, \{Y\}\}; T) > 0$
- Positive synergistic information: $\Pi_R(\{\{X, Y\}\}; T) > 0$
- Zero unique information: $\Pi_R(\{\{X\}\}; T) = 0$ and $\Pi_R(\{\{Y\}\}; T) = 0$

In such cases, we would have:

$$I(X; T) = \Pi_R(\{\{X\}, \{Y\}\}; T) > 0 \quad (3.42)$$

$$I(X; T \mid Y) = \Pi_R(\{\{X, Y\}\}; T) > 0 \quad (3.43)$$

This means X always has positive mutual information with T , both unconditionally and when conditioning on Y , yet provides no unique information. The Persistent Relevance Property rules out such counterintuitive scenarios by requiring that persistent relevance (positive conditional MI under all possible conditioning) must imply unique information contribution.

Definition 3.2.2 (Unique Information Shorthand). For a set of variables V with target variable $X_k \in V$ and source variable $X_i \in V \setminus \{X_k\}$, we define the shorthand notation:

$$\Pi_R^{V'}(\{\{X_i\}\}; X_k) \equiv \Pi_R^{\{X_i, X'\}}(\{\{X_i\}\}; X_k) \quad (3.44)$$

where:

- $X' = V \setminus \{X_i, X_k\}$ is the supervariable containing all variables except X_i and the target X_k
- $V' = \{X_i, X'\}$ is the bivariate source set for the PID decomposition

This notation represents the unique information that variable X_i provides about target X_k that cannot be obtained from any combination of all other variables in V .

Remark 3.2.4 (Interpretation of Unique Information Shorthand). The shorthand $\Pi_R^{V'}(\{\{X_i\}\}; X_k)$ captures the irreducible contribution of variable X_i about target X_k in the context of all available variables V . By grouping all other variables into the supervariable X' , we ensure that the unique information represents only what X_i can provide that is truly unavailable from any combination of the remaining variables. This formulation maintains the theoretical guarantees of bivariate PID while enabling analysis of the unique contributions of individual variables within larger variable sets.

3.2.1 Direct causal neighbors and unique information

We now introduce our central result establishing the correspondence between unique information and causal parents and children. Intuitively, the direct causal parents and children of a variable should provide unique information about that variable: a parent should exert an influence on the variable that cannot be derived from any other source, while children should be influenced by the variable in ways that reflect information unavailable elsewhere in the system. Our result formalizes this intuition and introduces appropriate theoretical assumptions to ensure that the correspondence between unique information and direct causal neighbors is bidirectional.

Theorem 3.2.2 (Causal Relationships and Unique Information). *Given a probability distribution that is faithful to a causal DAG and satisfies persistent relevance, for any variable X_i , another variable X_j has unique information about X_i if and only if X_j is a causal parent or child of X_i .*

Formally: $\Pi_R^{V'}(\{\{X_j\}\}; X_i) > 0 \Leftrightarrow X_j \in \text{pa}(X_i) \cup \text{ch}(X_i)$

Proof. Direction 1 ($\Rightarrow$): If X_j is not a causal parent or child of X_i , then $\Pi_R^{V'}(\{\{X_j\}\}; X_i) = 0$.

We consider two cases:

Case 1: $X_j \notin \text{MB}(X_i)$ (not in the Markov blanket (Def 2.2.18) of X_i).

By the Markov property, variables outside the Markov blanket are conditionally independent of X_i given the Markov blanket:

$$X_j \perp X_i \mid \text{MB}(X_i) \quad (3.45)$$

This gives $I(X_j; X_i \mid \text{MB}(X_i)) = 0$. In the bivariate PID decomposition with source variables $\{X_j, \text{MB}(X_i)\}$ (treating the Markov blanket as a single variable), this implies:

$$\Pi_R^{\{X_j, \text{MB}(X_i)\}}(\{\{X_j\}\}; X_i) = 0 \quad (3.46)$$

Since $\text{MB}(X_i) \subseteq V' = V \setminus \{X_i, X_j\}$, by the unique information monotonicity property (Definition 3.2.1):

$$Pi_R^{V'}(\{\{X_j\}\}; X_i) \leq \Pi_R^{\{X_j, \text{MB}(X_i)\}}(\{\{X_j\}\}; X_i) = 0 \quad (3.47)$$

Therefore, $\Pi_R^{V'}(\{\{X_j\}\}; X_i) = 0$.

Case 2: $X_j \in MB(X_i)$ but $X_j \notin pa(X_i) \cup ch(X_i)$.

Then X_j must be a co-parent of some child of X_i . Let $X_k \in ch(X_i) \cap ch(X_j)$ be a common child of X_i and X_j , so we have the V-structure $X_i \rightarrow X_k \leftarrow X_j$.

We can construct a conditioning set C as follows:

- Do not include X_k in C (to avoid activating the collider)
- For any other path between X_i and X_j , include appropriate variables in C to block those paths

By d-separation, with this choice of C :

$$X_j \perp X_i \mid C \quad (3.48)$$

Since $C \subseteq V \setminus \{X_i, X_j\} = V'$, by the unique information monotonicity property:

$$\Pi_R^{V'}(\{\{X_j\}\}; X_i) \leq \Pi_R^{\{X_j, C\}}(\{\{X_j\}\}; X_i) = 0 \quad (3.49)$$

Therefore, $\Pi_R^{V'}(\{\{X_j\}\}; X_i) = 0$.

Direction 2 ($\Leftarrow$): If X_j is a causal parent or child of X_i , then $\Pi_{\mathbf{R}}(\{\{X_j\}\}; X_i) > 0$.

If X_j is a causal parent or child of X_i , then there is a direct causal edge between them. By faithfulness (See Def. 2.2.10), this direct causal connection implies $I(X_j; X_i \mid C) > 0$ for all possible conditioning sets $C \subseteq S \setminus \{X_j\}$. By the persistent relevance property, this implies $\Pi_{\mathbf{R}}(\{\{X_j\}\}; X_i) > 0$. $\square$

Remark 3.2.5 (Localist Causal Discovery via Unique Information). The connection between unique information and causal parents and children exemplifies our localist interpretation of causal roles. Instead of narrowing down causal models through systematic conditional independence queries across the entire variable set, we can focus on any particular variable of interest and identify those variables that contribute unique information to it. Under the appropriate theoretical conditions, these variables that provide unique information will be exactly the causal parents and children of the target variable. This approach translates causal discovery from a global identification perspective over the space of graphical models into a local information-theoretic analysis centered on individual variables and their immediate causal neighborhoods.

3.2.2 Co-parent and synergy

After identifying the PID signature of causal parents and children, we seek to distinguish between these two roles to obtain a more complete causal characterization. The most reliable approach for making this distinction is through the identification of co-parent variables, which form collider structures with the target variable. Indeed, the collider structure represents the primary mechanism through which we can determine causal direction between variables, as chain and confounder structures belong to the same Markov equivalence class despite having different causal directions. This makes co-parent identification a crucial component of any causal discovery procedure aimed at resolving directional ambiguities.

By characterizing the PID signature of co-parents, we will obtain a complete information-theoretic picture of a variable's Markov blanket, along with the specific causal role that each variable within that blanket plays. This comprehensive characterization enables us to distinguish

between parents (which provide unique information), children (which also provide unique information but can be identified through their co-parent relationships), and co-parents (which exhibit distinctive synergistic information patterns with the children they share).

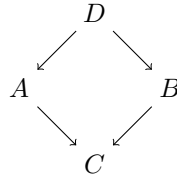
In this section, we identify the characteristic PID signature of causal co-parents and introduce appropriate theoretical assumptions to ensure that this identification is bidirectional—that is, co-parent relationships both imply and are implied by specific PID patterns under suitable conditions.

Assumption 3.2.3 (Collider Amplification Property). A joint probability distribution $P(X_i, X_j, X_k)$ satisfies the collider amplification property if for every V-structure $X_i \rightarrow X_j \leftarrow X_k$ in the underlying causal graph:

$$I(X_i; X_k \mid X_j) > I(X_i; X_k) \quad (3.50)$$

In other words, conditioning on the collider variable X_j strictly increases the mutual information between its co-parents X_i and X_k compared to their unconditional mutual information.

Remark 3.2.6 (Significance of Collider Amplification). Although the pure V-structure satisfies $I(X_i; X_k) = 0$ by d-separation, there may be additional dependencies between X_i and X_k induced by other variables in the graph, resulting in $I(X_i; X_k) > 0$. When this happens, d-separation principles alone do not specify the relative magnitude of $I(X_i; X_k)$ versus $I(X_i; X_k \mid X_j)$. For instance, if we consider a diamond structure like the following:



Then by d-separation, A and B will both be unconditionally dependent due to the confounder D , and conditionally dependent given C , but this alone doesn't tell us whether the mutual information between A and B will increase when the collider C is conditioned upon.

The collider amplification property formalizes the intuition that the collider effect should increase mutual information between co-parents, regardless of whether they have nonzero unconditional mutual information beforehand.

Definition 3.2.3 (Information Monotonicity Property). A redundancy measure $I_\cap$ satisfies the information monotonicity property if the following holds: Let S be a collection of source variables, $X \in S$, and C be another collection of random variables with $C \not\subset S$. Let $X' = \{X\} \cup C$ be the variable that combines X and C , and let S' be the source set obtained by replacing X in S with X' . Then for all lattice elements α in the redundancy lattice of S that contain X , let α' be the corresponding element obtained by replacing X with X' . We require:

$$\Pi_R^{S'}(\alpha') \geq \Pi_R^S(\alpha) \quad (3.51)$$

Remark 3.2.7 (Significance of Information Monotonicity). The information monotonicity property formalizes the intuitive notion that expanding a variable's information content cannot decrease any information-theoretic component involving that variable. Since $X' = \{X\} \cup C$ contains all the information that X has (and potentially more from C), replacing X with X' in any lattice element should preserve or increase the corresponding PID component.

This property ensures that when we replace X with the augmented variable X' , any synergistic, redundant, or unique information component will maintain access to the original information in X while potentially gaining additional information from C .

Theorem 3.2.3 (Co-parent Detection via Synergistic Information). *Given a collection of variables V and a joint probability distribution P that satisfies faithfulness, persistent relevance, and collider amplification, a variable $X_k \in V$ is a co-parent of variable $X_i \in V$ (i.e., there exists $X_j \in V$ such that $X_k \rightarrow X_j \leftarrow X_i$) if and only if:*

1. X_k has zero unique information about X_i : $\Pi_R^{V'}(\{\{X_k\}\}; X_i) = 0$
2. There exists $X_j \in V$ with $\Pi_R^V(\{\{X_j\}\}) > 0$ such that X_k and X_j have positive synergistic information about X_i : $\Pi_R^{\{X_j, X_k\}}(\{\{X_j, X_k\}\}; X_i) > 0$, where X_j is the joint child of X_k and X_i .

Note that the 0 unique information is relative to the set of all variables V , while positive synergy is only relative to $\{X_j, X_k\}$

Proof. Direction 1 ($\Leftarrow$): If X_k is a co-parent of X_i , then the conditions hold.

Zero unique information: Since X_k is not a direct parent or child of X_i (but rather connected through the collider X_j), by Theorem 3.2.2, we have $\Pi_R^{V'}(\{\{X_k\}\}; X_i) = 0$.

Positive synergistic information: Consider the V-structure $X_k \rightarrow X_j \leftarrow X_i$, since X_j is a child of X_i , we have $\Pi_R(\{\{X_j\}\}) > 0$. By the collider amplification property:

$$I(X_k; X_i \mid X_j) > I(X_k; X_i) \quad (3.52)$$

From the two-source PID decomposition:

$$I(X_k; X_i \mid X_j) = \Pi_R^{\{X_j, X_k\}}(\{\{X_k\}\}; X_i) + \Pi_R^{\{X_j, X_k\}}(\{\{X_j, X_k\}\}; X_i) \quad (3.53)$$

$$I(X_k; X_i) = \Pi_R^{\{X_j, X_k\}}(\{\{X_k\}, \{X_j\}\}; X_i) + \Pi_R^{\{X_j, X_k\}}(\{\{X_k\}\}; X_i) \quad (3.54)$$

Substituting into the collider amplification inequality:

$$\Pi_R^{\{X_j, X_k\}}(\{\{X_j, X_k\}\}; X_i) - \Pi_R^{\{X_j, X_k\}}(\{\{X_k\}, \{X_j\}\}; X_i) > 0 \quad (3.55)$$

Since redundant information is non-negative, $\Pi_R^{\{X_j, X_k\}}(\{\{X_k\}, \{X_j\}\}; X_i) \geq 0$, we conclude:

$$\Pi_R^{\{X_j, X_k\}}(\{\{X_j, X_k\}\}; X_i) > 0 \quad (3.56)$$

Direction 2 ($\Rightarrow$): If the conditions hold, then X_k is a co-parent of X_i .

We prove this by contradiction. Assume that variable X_k has zero unique information about X_i but is not a co-parent of X_i . We will show that there cannot exist any variable X_j with positive unique information such that X_k and X_j have positive synergistic information about X_i .

Since X_k has zero unique information about X_i , by Theorem 3.2.2, X_k is not a causal parent or child of X_i . Combined with our assumption that X_k is not a co-parent, this implies that X_k is outside the Markov blanket of X_i : $X_k \notin MB(X_i)$.

On the other hand, any variable X_j with positive unique information about X_i must be a causal parent or child of X_i by Theorem 3.2.2, which means $X_j \in MB(X_i)$.

By the Markov property, since $X_k \notin MB(X_i)$:

$$I(X_k; X_i \mid MB(X_i)) = 0 \quad (3.57)$$

This implies that in the bivariate PID decomposition with source variables $\{X_k, MB(X_i)\}$ (treating the Markov blanket as a single variable):

$$\Pi_R^{\{X_k, MB(X_i)\}}(\{\{X_k, MB(X_i)\}\}; X_i) = 0 \quad (3.58)$$

Since $X_j \in MB(X_i)$, by the information monotonicity property (Definition 3.2.3), we have:

$$\Pi_R^{\{X_k, MB(X_i)\}}(\{\{X_k, MB(X_i)\}\}; X_i) \geq \Pi_R^{\{X_k, X_j\}}(\{\{X_k, X_j\}\}; X_i) \quad (3.59)$$

Combining equations 3.58 and 3.59, we have:

$$\Pi_R^{\{X_k, X_j\}}(\{\{X_k, X_j\}\}; X_i) = 0 \quad (3.60)$$

Therefore, X_k and X_j have zero bivariate synergistic information, contradicting the assumption that there exists positive synergistic information. Hence, X_k must be a co-parent of X_i . $\square$

Remark 3.2.8 (Directional Confirmation via Co-parent Identification). Identifying the PID signature of a co-parent enables us to confirm whether a variable B is a child of the target variable A . Conversely, if we want to verify that B is a parent of A , we can reverse the roles by setting B as the target and attempting to identify a co-parent of B with A serving as their joint child. This bidirectional verification process aligns with our “localist” motivation, where we systematically characterize the profile of each variable’s immediate causal neighborhood.

After identifying the causal neighborhood of every variable through this localist approach, we can piece together a global picture of the Bayesian network by integrating the local structural information. When no co-parent relationship exists to confirm whether B is a parent or child of A , we may need to resort to additional constraints or heuristics to determine the causal direction, such as imposing acyclicity constraints to avoid the formation of cycles. Alternatively, it may be the case that the causal direction remains fundamentally indeterminate within the Markov equivalence class that our data can identify, reflecting the inherent limitations of observational causal discovery even with information-theoretic approaches.

Theorem 3.2.4 (PID-Based Bayesian Network Causal Discovery Procedure). *Given a set of variables V and a joint probability distribution that satisfies faithfulness, persistent relevance, and appropriate PID properties, the following three-step procedure identifies the causal structure of the underlying Bayesian network:*

Step 1: Direct Causal Neighbor Identification. *For each variable $X_i \in V$, set X_i as the target for PID analysis and identify all variables with positive unique information about X_i :*

$$N(X_i) = \{X_j \in V \setminus \{X_i\} : \Pi_R^{V'}(\{\{X_j\}\}; X_i) > 0\}$$

By Theorem 3.2.2, $N(X_i) = pa(X_i) \cup ch(X_i)$ represents the direct causal neighbors of X_i .

Step 2: Directional Confirmation via Co-parent Detection. *For each variable $X_i \in V$ and each of its direct causal neighbors $X_j \in N(X_i)$, attempt to identify a co-parent variable X_k using Theorem 3.2.3. Specifically, search for variables $X_k \in V$ such that:*

- $\Pi_R^{V'}(\{\{X_k\}\}; X_i) = 0$ (zero unique information about X_i)
- $\Pi_R^{\{X_j, X_k\}}(\{\{X_j, X_k\}\}; X_i) > 0$ (positive synergistic information with X_j)

The identification of such co-parent X_k confirms that $X_j \in ch(X_i)$ (equivalently, $X_i \in pa(X_j)$), thereby establishing the causal direction $X_i \rightarrow X_j$.

Step 3: Remaining Direction Resolution. *After all identifiable co-parent relationships have been used to direct edges, apply remaining heuristics such as acyclicity constraints to complete any indeterminate causal relationships between variables that could not be resolved through co-parent identification.*

This procedure systematically leverages the localist nature of PID signatures to construct a global causal model while respecting the theoretical limitations of information-theoretic causal discovery.

3.2.3 PID type signature in Bayesian Hypergraphs

After identifying the PID signatures of the primary causal roles in Bayesian networks, we now extend this approach to Bayesian hypergraphs. One significant difference in the expressivity of

Bayesian hypergraphs is that while the collider effect in standard Bayesian networks induces conditional dependence among all parents of a variable when conditioned upon, in Bayesian hypergraphs the collider effect activates only partially—specifically among variables that are co-tails within the same hyperedge. This fundamental difference will be reflected in the synergistic information behavior among the parents of a variable, which forms our theoretical basis for extending PID-based causal discovery to the hypergraph setting.

Another key advantage of hypergraphs is that the co-head relationship provides a natural framework for accommodating indeterminacy in causal direction between variables. When unconditional tests reveal dependencies without clear directional information, hypergraphs can represent these relationships through co-head structures that capture the symmetric aspects of the interaction while preserving the overall causal hierarchy.

In our hypergraph extension, we identify the characteristic PID signatures of variables within individual hyperedges, capturing both the directed relationships between tails and heads and the symmetric relationships among co-heads and co-tails. To reverse this process for causal discovery, we systematically search for subsets of variables that satisfy the PID signatures characteristic of hyperedge structures. We then select the “maximal” sets that match these signatures—that is, sets which would no longer satisfy the hyperedge signature if expanded further. This maximal hyperedge construction allows us to resolve ambiguities among a wide range of Markov equivalent models by selecting a canonical representative of the equivalence class.

Definition 3.2.4 (Hypergraph Collider Amplification Property). A joint probability distribution P that factorizes according to a directed acyclic hypergraph $D = (V, E, \phi)$ satisfies the hypergraph collider amplification property if for any directed hyperedge $h \in E$ and any two distinct variables $X_i, X_k \in T(h)$ (cotail variables) and any variable $X_j \in H(h)$ (head variable):

$$I(X_i; X_k \mid X_j) > I(X_i; X_k) \quad (3.61)$$

In other words, conditioning on any head variable X_j strictly increases the mutual information between cotail variables X_i and X_k compared to their unconditional mutual information.

Remark 3.2.9 (Significance of Cotail Restriction). The hypergraph collider amplification property is crucial because the collider effect only activates for variables that are cotails within the same directed hyperedge. This restriction is necessary due to the factorization structure of directed hypergraphs.

Consider a hypergraph with three variables X_1, X_2, X_3 and two directed hyperedges:

- e_1 with $T(e_1) = \{X_1\}$ and $H(e_1) = \{X_2\}$
- e_2 with $T(e_2) = \{X_3\}$ and $H(e_2) = \{X_2\}$

The factorization becomes:

$$f(x_1, x_2, x_3) = \psi_{12}(x_1, x_2)\psi_{32}(x_3, x_2) \quad (3.62)$$

In this case, X_1 and X_3 are not cotails of any single hyperedge. By the Markov properties of the resulting undirected hypergraph structure, conditioning on X_2 does not create conditional dependence between X_1 and X_3 , since they belong to separate potential functions that only share the variable X_2 . The collider amplification effect requires variables to be joint causes (cotails) of the same effect (head) within a single directed hyperedge.

Theorem 3.2.5 (Hypergraph Unique Information Characterization). *Given a probability distribution that is faithful to a directed acyclic hypergraph $D = (V, E, \phi)$ and satisfies the persistent relevance property, for any variable $X_i \in V$, another variable $X_j \in V$ has positive unique information about X_i if and only if X_j is a parent, child, or cohead of X_i .*

Formally: $\Pi_R^{V'}(\{\{X_j\}\}; X_i) > 0 \Leftrightarrow X_j \in \text{pa}(X_i) \cup \text{ch}(X_i) \cup \{X_k : X_k \sim_H X_i\}$

where $X_k \sim_H X_i$ denotes that X_k and X_i are coheads in some directed hyperedge.

Proof Sketch. We provide a proof sketch as the proof is similar to that of theorem 3.2.2:

Direction 1 ($\Leftarrow$): If X_j is a parent, child, or cohead of X_i , then $\Pi_R^{V'}(\{\{X_j\}\}; X_i) > 0$.

Parents and children: By the same argument as in 3.2.2, direct causal connections ensure $I(X_j; X_i \mid C) > 0$ for all conditioning sets $C \subseteq V \setminus \{X_j\}$. By persistent relevance, this implies positive unique information.

Coheads: If $X_j \sim_H X_i$, they appear together in the head $H(h)$ of some directed hyperedge h . By the hypergraph factorization structure, this creates a direct statistical dependency through the shared potential function ψ_h that cannot be eliminated by conditioning on other variables, ensuring $I(X_j; X_i \mid C) > 0$ for all $C \subseteq V \setminus \{X_j\}$.

Direction 2 ($\Rightarrow$): If X_j is not a parent, child, or cohead of X_i , then $\Pi_R^{V'}(\{\{X_j\}\}; X_i) = 0$.

Outside Markov blanket: If $X_j \notin MB(X_i)$, then by the Markov property: $X_j \perp X_i \mid MB(X_i)$, giving $I(X_j; X_i \mid MB(X_i)) = 0$. By the argument in Theorem 3.2.2, this implies zero unique information.

Cotails: If X_j is a cotail of X_i (i.e., $X_j \sim_T X_i$ but not cohead), then there exists a conditioning set C that excludes their common head variables but includes appropriate blocking variables for other paths. By the hypergraph Markov properties, this renders $X_j \perp X_i \mid C$, yielding $I(X_j; X_i \mid C) = 0$ and thus zero unique information. $\square$

Theorem 3.2.6 (Hypergraph Co-tail Detection via Synergistic Information). *Given a probability distribution that is faithful to a directed acyclic hypergraph $D = (V, E, \phi)$ and satisfies persistent relevance and hypergraph collider amplification, a variable $X_k \in V$ is a co-tail of variable $X_i \in V$ (i.e., there exists a directed hyperedge h such that $X_k, X_i \in T(h)$) if and only if:*

1. X_k has zero multivariate unique information about X_i : $\Pi_R^{V'}(\{\{X_k\}\}; X_i) = 0$
2. There exists a variable $X_j \in V$ such that:
 - X_j has positive multivariate unique information about X_i : $\Pi_R^{V'}(\{\{X_j\}\}; X_i) > 0$
 - X_k and X_j have positive bivariate synergistic information about X_i : $\Pi_R^{\{X_j, X_k\}}(\{\{X_j, X_k\}\}; X_i) > 0$

Proof Sketch. **Direction 1** ($\Leftarrow$): If X_k is a co-tail of X_i , then the conditions hold.

Zero multivariate unique information: Since X_k is not a parent, child, or cohead of X_i (but rather a co-tail), by the previous theorem, we have $\Pi_R^{V'}(\{\{X_k\}\}; X_i) = 0$.

Positive synergistic information: Consider the directed hyperedge h with $X_k, X_i \in T(h)$ and $X_j \in H(h)$. By the hypergraph collider amplification property:

$$I(X_k; X_i \mid X_j) > I(X_k; X_i) \quad (3.63)$$

Using the same PID decomposition argument as in Theorem 2.3.2, this inequality implies:

$$\Pi_R^{\{X_j, X_k\}}(\{\{X_j, X_k\}\}; X_i) > 0 \quad (3.64)$$

Since X_j is in the head of the hyperedge containing X_i in its tail, X_j is a child of X_i , ensuring $\Pi_R^{V'}(\{\{X_j\}\}; X_i) > 0$ by the previous theorem.

Direction 2 ($\Rightarrow$): If the conditions hold, then X_k is a co-tail of X_i .

We proceed by contradiction similar to Theorem 3.2.3. The key distinction is that the Markov blanket structure in hypergraphs differs from ordinary graphs. If X_k is not a co-tail of X_i , then even if they share common head variables, they belong to separate hyperedges in the factorization.

As demonstrated in Remark 3.2.9, variables that are not co-tails can be conditionally independent even when they share the same head variable, since they appear in separate potential functions. This conditional independence, combined with the information monotonicity property, leads to zero bivariate synergistic information, contradicting the assumed positive synergy. Therefore, X_k must be a co-tail of X_i . $\square$

Theorem 3.2.7 (PID Signature of Hyperedges). *Given a probability distribution that is faithful to a directed acyclic hypergraph $D = (V, E, \phi)$ and satisfies persistent relevance and hypergraph collider amplification, for any directed hyperedge $h \in E$ with head $H(h)$ and tail $T(h)$, the following PID signature properties hold:*

1. For every $X_i \in T(h)$ and $X_j \in H(h)$, any co-tail $X_k \in T(h) \setminus \{X_i\}$ has positive bivariate synergistic information with X_j about X_i :

$$\Pi_R^{\{X_j, X_k\}}(\{\{X_j, X_k\}\}; X_i) > 0 \quad (3.65)$$

2. For every $X_i \in T(h)$ and $X_j \in H(h)$, the head variable X_j has positive multivariate unique information about X_i :

$$\Pi_R^{V'}(\{\{X_j\}\}; X_i) > 0 \quad (3.66)$$

3. For any pair of coheads $X_j, X_k \in H(h)$ with $X_j \neq X_k$, they have positive multivariate unique information about each other:

$$\Pi_R^{V'}(\{\{X_j\}\}; X_k) > 0 \text{ and } \Pi_R^{V'}(\{\{X_k\}\}; X_j) > 0 \quad (3.67)$$

Proof Sketch. The theorem follows directly from our previous results:

Property 1: By Theorem 3.2.6, co-tail variables X_k and X_i exhibit positive bivariate synergistic information with their common head X_j .

Property 2: By Theorem 3.2.5, head variables X_j are children of tail variables X_i , ensuring positive unique information.

Property 3: By Theorem 3.2.5, co-head variables are directly connected through their shared hyperedge membership, guaranteeing mutual unique information. $\square$

Definition 3.2.5 (Maximal Hyperedge). A directed hyperedge $h^* \in E$ is called maximal if extending either its head $H(h^*)$ or tail $T(h^*)$ would cause it to violate the PID signature properties of Theorem 3.2.7.

Formally, h^* is maximal if there does not exist a hyperedge h' with either:

- $T(h^*) \subset T(h')$ and $H(h^*) = H(h')$, or
- $T(h^*) = T(h')$ and $H(h^*) \subset H(h')$

such that h' also satisfies the PID signature properties.

Remark 3.2.10 (Hyperedge Consolidation). The maximal hyperedge construction provides a canonical representation that consolidates redundant hyperedge specifications. For example, consider three hyperedges:

- e_1 : $T(e_1) = \{X_1, X_2\}$, $H(e_1) = \{X_3\}$

- e_2 : $T(e_2) = \{X_1, X_2\}$, $H(e_2) = \{X_4\}$
- e_3 : $T(e_3) = \emptyset$, $H(e_3) = \{X_3, X_4\}$

These can be absorbed into a single maximal hyperedge: e^* with $T(e^*) = \{X_1, X_2\}$ and $H(e^*) = \{X_3, X_4\}$, which results in the same conditional independence properties while providing a more parsimonious representation of the hypergraph structure.

Building upon the definition of maximal hyperedges and the PID signature characterization of hyperedge structures, we present a systematic procedure for Bayesian hypergraph causal discovery that leverages information-theoretic patterns to identify canonical causal representations.

Theorem 3.2.8 (PID-Based Bayesian Hypergraph Causal Discovery Procedure). *Given a set of variables V and a joint probability distribution that satisfies faithfulness, persistent relevance, hypergraph collider amplification, and appropriate PID properties, the following procedure identifies the (maximal) bayesian hypergraph structure:*

Hyperedge Discovery Algorithm:

1. **Candidate Search:** Systematically search for subsets $T \subseteq V$ (potential tails) and $H \subseteq V$ (potential heads) with $T \cap H = \emptyset$ that satisfy the hyperedge PID signature properties of Theorem 3.2.7.
2. **Maximal Extension:** For each candidate hyperedge (T, H) that satisfies the PID signature, expand both the tail set T and head set H by adding variables while maintaining the hyperedge signature properties, continuing until no further expansion is possible without violating the signature.
3. **Canonical Selection and Iterative Construction:** Select the maximal hyperedge (T^*, H^*) that cannot be extended in either tail or head without violating the PID signature properties, as defined in Definition 3.2.5. Continue this search process iteratively until no additional maximal hyperedges can be identified.

By consistently selecting the maximal hyperedge that satisfies our PID signature properties, we establish a systematic approach to resolving the ambiguity inherent in Markov equivalent hypergraph models through a clear canonical selection rule. As previously discussed, hypergraphs provide the representational flexibility to model probability distributions where the collider effect activates only within co-tail relationships, while co-head relationships naturally accommodate causal interactions with indeterminate directional information.

Our theoretical results translate these structural properties into distinctive patterns in the synergistic and unique information contributions of the constituent variables. By systematically searching for these information-theoretic patterns in empirical data, we can identify the hyperedge structures surrounding any particular variable of interest. The localist nature of these PID signatures enables us to characterize the causal behavior and neighborhood structure of individual variables without requiring global identification of the entire hypergraph structure—a significant computational and inferential advantage for practical causal discovery applications.

4

Discussion

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4.1 Summary and conclusions

As outlined in the introduction, our motivation for this thesis was twofold: first, to discover the limits of unconditional tests for causal discovery and, given these constraints, to leverage unconditional tests to their fullest extent while respecting theoretical boundaries; second, to advance beyond unconditional interaction measures by utilizing targeted partial information decomposition for causal discovery, identifying the PID signatures that characterize major causal roles such as parents, children, and co-parents.

For our first objective, we began by providing a rigorous proof of the known technique for V-structure detection using Lancaster interaction measures. We then conducted a systematic analysis of the precise limits of unconditional tests in three-variable structures and their extensions, which led us to define the notion of unconditional independence equivalence classes. This theoretical framework provides a principled coarse-graining that crisply delineates the boundary between causal roles that can and cannot be identified through unconditional tests. This equivalence class structure serves as the foundation for our procedure that leverages Bayesian hypergraphs to identify causal structures using unconditional tests, thereby utilizing their power to the full theoretical limits while acknowledging inherent constraints.

For our second objective, we established the theoretical foundation connecting PID to causality through the key insight that knowledge of all PID components enables recovery of all conditional mutual information terms, providing complete information about the conditional independence relationships used to identify graphical models. Building on this motivation, we characterized the distinctive PID signatures of direct causal neighbors (parents and children) as well as co-parents, enabling systematic distinction between parental and child relationships. These results culminate in a comprehensive procedure for causal discovery using partial information decomposition.

Our PID signature results establish the foundation for a “localist interpretation” of causality. Since PID signatures are defined relative to a target variable, researchers interested in only a

subset of variables can focus their analysis on specific parts of the causal graph, examining the direct causal neighborhood and identifying factors that directly influence or are influenced by variables of interest, without requiring identification of the entire graph structure. This localist approach offers both computational advantages and interpretive clarity for understanding specific causal relationships.

We extended this framework to Bayesian hypergraphs by characterizing the PID signature of hyperedges, enabling representation of more fine-grained relationships such as partial activation of collider effects and accommodation of ambiguous causal relationships through co-head structures. These enhanced expressive capabilities are reflected in distinctive synergy patterns between variables and align naturally with our localist analytical framework, providing tools for understanding complex causal mechanisms that cannot be adequately captured by traditional pairwise graphical models.

Through these contributions, we have established information-theoretic foundations for causal discovery that complement traditional conditional independence approaches, providing both theoretical insights into fundamental limits and practical procedures for extracting maximal causal information within those constraints.

4.2 Limitations

For our first motivation of using unconditional tests for causal discovery, discovering the limitations of unconditional tests is actually one of our main research objectives, and hence these constraints have been thoroughly discussed throughout the body of the thesis. The equivalence class framework we developed precisely characterizes these boundaries, showing which causal roles can and cannot be distinguished through unconditional information alone.

For our second motivation of using PID for causal discovery, several important limitations constrain the applicability of our approach. First, knowledge of all PID components is fundamentally limited by the power of conditional independence statements, since PID components enable recovery of conditional mutual information terms, which vanish exactly when conditional independence holds. This means that certain causal structures can only be determined through interventional experiments and not through our PID signatures, simply because these structures cannot be distinguished through observational conditional independence queries alone. Most notably, chain and confounder structures belong to the same Markov equivalence class and therefore cannot be distinguished through conditional independence tests or our PID-based approach.

Another fundamental limitation arises from our faithfulness assumption, which presupposes that the conditional independence properties of the underlying joint probability distribution can be fully characterized by some Bayesian network or hypergraph structure. However, many probability distributions cannot be faithfully represented by Bayesian networks or hypergraphs, since the conditional independence properties of joint distributions are constrained only by the (semi)graphoid axioms, which are considerably more permissive than structural criteria such as d-separation. We present examples of these representational limitations in the following:

Example 4.2.1 (Limitations of the Faithfulness Assumption). The faithfulness assumption, while mathematically convenient, can fail in practice when deterministic relationships are present. A classic counterexample is the XOR relationship [9].

Consider three binary variables $X, Y, Z \in \{0, 1\}$ with the deterministic relationship $Z = X \oplus Y$ (XOR), where all eight combinations of (X, Y, Z) satisfying this constraint are equally likely. This distribution exhibits the following conditional independence pattern:

$$X \perp Y \perp Z \quad (\text{pairwise marginal independence}) \quad (4.1)$$

$$X \not\perp Y \mid Z, \quad Y \not\perp Z \mid X, \quad X \not\perp Z \mid Y \quad (\text{conditional dependence}) \quad (4.2)$$

This independence structure cannot be faithfully represented by any directed acyclic graph. Every DAG that captures the three-way conditional dependencies will necessarily violate the pairwise marginal independencies, and vice versa.

The fundamental issue arises because Bayesian networks satisfy the graphoid axioms (Def 2.2.11), which assume strictly positive probability densities. Deterministic relationships violate this assumption by assigning zero probability to certain configurations (e.g., $P(Z = 0, X = 1, Y = 1) = 0$ in the XOR case).

Example 4.2.2 (Limitations of Hypergraph Representation). While directed hypergraphs extend the representational power of standard Bayesian networks, they still face fundamental limitations in capturing certain complex causal mechanisms. In a directed hypergraph, if we have a hyperedge e with head $H(e)$ and tail $T(e)$, then under faithfulness, conditioning on any variable in the head should induce conditional dependence between the tail variables. However, we can construct scenarios where this property fails.

Consider the following counterexample: Let A and B be independent coin flips, and define $C = A \vee B$ (logical OR), making C a collider for A and B . Let C_1 be another independent coin flip, and define C_2 deterministically as:

$$C_2 = \begin{cases} C_1 & \text{if } C = 1 \\ \neg C_1 & \text{if } C = 0 \end{cases} \quad (4.3)$$

Suppose that only variables A, B, C_1, C_2 are observable while C is hidden. In this scenario:

- All four observable variables are marginally independent
- Conditioning on C_1 alone does not induce conditional dependence between A and B (since C_1 is an independent coin flip)
- Conditioning on C_2 alone does not induce conditional dependence between A and B
- Conditioning on both C_1 and C_2 jointly allows deduction of C , which then induces conditional dependence between A and B via the collider effect

This pattern cannot be expressed by a Bayesian hypergraph, since any hyperedge representation would require that conditioning on individual head variables induces the collider effect under faithfulness. Our counterexample represents a case where two variables A and B jointly determine the relationship between C_1 and C_2 , highlighting a fundamental limitation where the causal influence operates through complex logical combinations rather than simple potential functions over hyperedge variables.

Finally, while our PID approach circumvents the combinatorial explosion typical in multivariate PID by transforming to a bivariate framework via construction of supervariables, there may still be significant computational costs associated with working with PID measures in general. For instance, the BROJA measure defined in [6] quantifies unique information through a constrained optimization over the space of all probability distributions that admit particular marginals, which results in a high-dimensional optimization problem even when only three variables are involved. These computational challenges may limit the practical applicability of our theoretical results to large-scale causal discovery problems, where the supervariable construction itself may involve high-dimensional joint distributions that are difficult to estimate accurately from finite data.

4.3 Future Work

While joint probability distributions provide an appropriate theoretical framework for our analysis, real-world applications require working with finite data samples rather than complete distributional knowledge. Kernel-based methods [1] grounded in reproducing kernel Hilbert spaces serve

as a bridge from theoretical probability distributions to practical implementation by translating statements about distributions into statements about expectations, which can be approximated empirically from data. One natural future direction is to adapt our hypergraph-based unconditional test causal discovery algorithm to utilize kernel-based methods for implementing unconditional tests in practice. This adaptation should be relatively straightforward given that kernel-based unconditional independence testing is an established technique in the machine learning literature.

This computational bridge is equally relevant to our PID-based results on causal role signatures. Similar to conditional independence tests, PID fundamentally operates on joint probability distributions to quantify unique, synergistic, and redundant information. A computationally efficient measure that estimates these information components from real-world data would establish the practical connection between our theoretical results on PID signatures and real-world causal discovery applications, further strengthening our “localist” agenda by enabling focused analysis of specific variables of interest without requiring complete distributional knowledge.

One major limitation of our PID results stems from the faithfulness assumption underlying standard graphical models. A promising future direction involves investigating PID signatures of causal roles within more expressive causal frameworks than Bayesian networks or hypergraphs. Factored space models [10] represent one such framework, forming semigraphoids rather than graphoids and removing positivity assumptions to better accommodate deterministic relationships. This enhanced expressivity may enable modelling the causality of abstractions from detailed microstates to coarse-grained macrostates. Exploring PID signatures within factored space models could improve the scope of our analysis, similar to how extending from Bayesian networks to hypergraphs increased representational power.

Another compelling direction involves extending our PID signature results to algorithmic information theory. The Bayesian network causality framework translates naturally to algorithmic settings through the algorithmic Markov condition [11], where finite strings replace random variables and causal arrows correspond to computational orderings that optimize compression of all involved strings. Algorithmic information theory characterizes object complexity through the length of the shortest program generating that object, formalizing Occam’s razor without reference to probability distributions. While PID has not been formalized in algorithmic information theory, we believe that with appropriate algorithmic PID measures, our results connecting PID to causality could translate to form bridges with algorithmic Markov condition characterizations of causality. This would advance our “localist” agenda toward a domain that employs intrinsic notions of simplicity to formalize causality without relying on underlying joint distributions.

Bibliography

- [1] D. Sejdinovic, A. Gretton, and W. Bergsma, “A kernel test for three-variable interactions,” in *Advances in Neural Information Processing Systems*, C. Burges, L. Bottou, M. Welling, Z. Ghahramani, and K. Weinberger, Eds., vol. 26, Curran Associates, Inc., 2013. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2013/file/076a0c97d09cf1a0ec3e19c7f25299Paper.pdf.
- [2] Z. Liu, M. Barahona, and R. Peach, “Information-theoretic measures on lattices for higher-order interactions,” in *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 258, PMLR, May 2025, pp. 2206–2214. [Online]. Available: <https://proceedings.mlr.press/v258/liu25f.html>.
- [3] C. M. Bishop, *Pattern Recognition and Machine Learning (Information Science and Statistics)*, 1st ed. Springer, 2007, ISBN: 0387310738. [Online]. Available: <http://www.amazon.com/Pattern-Recognition-Learning-Information-Statistics/dp/0387310738%3FSubscriptionId%3D13CT5CVB80YFWJEPWS02%26tag%3Dws%26linkCode%3Dxm2%26camp%3D2025%26creative%3D165953%26creativeASIN%3D0387310738>.
- [4] M. A. Javidian, L. Lu, M. Valtorta, and Z. Wang, *On a hypergraph probabilistic graphical model*, 2018. arXiv: [1811.08372](https://arxiv.org/abs/1811.08372) [cs.DS]. [Online]. Available: <https://arxiv.org/abs/1811.08372>.
- [5] P. L. Williams and R. D. Beer, *Nonnegative decomposition of multivariate information*, 2010. arXiv: [1004.2515](https://arxiv.org/abs/1004.2515) [cs.IT]. [Online]. Available: <https://arxiv.org/abs/1004.2515>.
- [6] N. Bertschinger, J. Rauh, E. Olbrich, J. Jost, and N. Ay, “Quantifying unique information,” *Entropy*, vol. 16, no. 4, pp. 2161–2183, 2014, ISSN: 1099-4300. DOI: [10.3390/e16042161](https://doi.org/10.3390/e16042161). [Online]. Available: <https://www.mdpi.com/1099-4300/16/4/2161>.
- [7] A. Lyu, A. Clark, and N. Raviv, *Multivariate partial information decomposition: Constructions, inconsistencies, and alternative measures*, 2025. arXiv: [2508.05530](https://arxiv.org/abs/2508.05530) [cs.IT]. [Online]. Available: <https://arxiv.org/abs/2508.05530>.
- [8] C. E. Shannon, “A mathematical theory of communication,” *The Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948. DOI: [10.1002/j.1538-7305.1948.tb01338.x](https://doi.org/10.1002/j.1538-7305.1948.tb01338.x).
- [9] T. S. Verma and J. Pearl, “On the equivalence of causal models,” *CoRR*, vol. abs/1304.1108, 2013. arXiv: [1304.1108](https://arxiv.org/abs/1304.1108). [Online]. Available: <http://arxiv.org/abs/1304.1108>.
- [10] S. Garrabrant, M. G. Mayer, M. Wache, L. Lang, S. Eisenstat, and H. Dell, *Factored space models: Towards causality between levels of abstraction*, 2024. arXiv: [2412.02579](https://arxiv.org/abs/2412.02579) [cs.AI]. [Online]. Available: <https://arxiv.org/abs/2412.02579>.
- [11] D. Janzing and B. Schoelkopf, *Causal inference using the algorithmic markov condition*, 2008. arXiv: [0804.3678](https://arxiv.org/abs/0804.3678) [math.ST]. [Online]. Available: <https://arxiv.org/abs/0804.3678>.